Forecasting Brazil's Presidential Runoff after the First Round: Calibrated Error Fusion in Logit Space

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Abstract

We forecast the second round of Brazil's 2026 presidential election (Flávio Bolsonaro × Lula) once the first-round result is known. Three estimates of the runoff two-party share are combined on the logit scale: a bridge from the official first-round result, modelled as a Markov transition from first-round choices (including blank/null votes and abstention) to runoff choices, with retention, defection and turnout flows from ecological inference on 2002–2022 and pollster transfer grids for the eliminated candidates; head-to-head runoff polls fielded before the first round, and head-to-head polls fielded after it. Error magnitudes are measured on data rather than set by hand: the bridge out of sample on Brazil's six runoffs of 2002–2022 (RMSE 0.168 logit), and the polls, by horizon, on runoffs in eight countries from the Jennings & Wlezien (2018) archive. Runoff polls are shifted by +1.3 pp towards the PT's opponent, the mean overstatement of the PT in the final runoff polls of all six elections. The first-round poll miss is not carried over (corr = 0.06 across 21 elections). The components share the opinion drift between the date of their information and election day, so they are combined by generalised least squares with covariance $\sigma_d^2 \min(h_i, h_j)$ instead of precision weighting, which counted that drift twice. In a leave-one-election-out backtest the bridge alone calls 6/6 winners on the day after the first round (share RMSE 4.1 pp); the structural component of earlier versions (prior plus government approval) worsens both the share error and the predictive density and is dropped. On 6 October 2026, with the TSE count at 100% (Flávio 47.03%, Lula 45.16%), the model gives Pr(Flávio) = 0.705 and an expected Flávio share of 52.2% (95% interval 44.1–60.3%). A separate study of the tail event, a comeback by the first-round runner-up that has never happened in Brazil, puts its probability at 0.26–0.32 and rejects base rates, international regressions and chaos-theoretic arguments as primary estimators. The superseded v6 framework is kept in Appendix B. Appendix C records the updates after the pre-registration: with 17 post-first-round polls from 13 pollsters (16 October), Pr(Flávio) = 0.875 and an expected share of 52.8% (48.0–57.6%), with no divergence alert.

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Disclosure regarding the use of artificial intelligence

This working paper was prepared with the assistance of generative artificial-intelligence tools (Anthropic Claude). The AI assisted in drafting expository text, writing and refactoring the data-processing and modelling code (Python), running the validation suites, suggesting references and editing the manuscript.

The methodological choices, modelling decisions, data interpretation and conclusions are the sole intellectual responsibility of the author, who has reviewed all AI-generated content, including derivations, code and numerical results, and assumes full responsibility for the accuracy and integrity of the paper. This disclosure follows the practice recommended by major journals (Nature 2023; Science 2023/2024; JASA editorial guidance).

Notation

Symbol	Meaning
s	Flávio's share of the runoff valid vote (two-party share)
$\varphi = \text{logit}(s)$	target on the logit scale; $\varphi > 0 \Leftrightarrow$ Flávio wins
$k \in \{B, H, F\}$	components: bridge, pre-first-round polls, post-first-round polls
$\mu_k,\ \sigma_k$	estimate of φ from component k and its calibrated RMSE
h_k	days between the information in component k and the runoff
$e_k = \mu_k - \varphi$	error of component k
σ_d	standard deviation per day of the common opinion drift
b	Brazilian runoff-poll bias, pp of two-party share in favour of the PT
V_c	first-round valid-vote share of candidate c
(f_c, l_c, u_c)	transfer grid: share of c's voters going to Flávio, to Lula, uncommitted
$\hat{m},\ \hat{\sigma}$	combined estimate of φ and its standard deviation
Φ	standard normal CDF

1. Introduction

Brazil elects its president in two rounds, and every runoff since 1989 has been won by the first-round leader. In 2026 the margin was the smallest on record: Flávio Bolsonaro (PL) 47.03%, Lula (PT) 45.16% of valid votes, with 7.8% for the other candidates. The question for the 25 October runoff is therefore narrow: how the remaining voters split, how far opinion moves in three weeks, and how much the polls can be trusted.

Versions up to 6.10 answered a different question. Built months before the first round, they combined runoff head-to-head polls with a structural prior driven by government approval, marginalised over who the right-wing candidate would be, and used variance components set by hand (Appendix B). Once the first round is known, three things change. The first-round result is the most informative single datum about the runoff. The structural component has to earn its place against it. And the error of each input can be measured rather than assumed, because poll and runoff archives exist for Brazil and for other two-round systems.

Version 7.0 is the result of that re-examination (7.1 adds abstention, blank/null votes and undecided voters to the bridge, §2.2), done one election at a time without letting the tested election inform its own calibration. It is deliberately simple: three estimates of the same quantity, each with an empirically measured error, combined in closed form. Section 2 states the model, Section 3 the data, Section 4 the validation, Section 5 what was tested and rejected, Section 6 the 2026 application and Section 7 the study of the comeback probability. Section 8 lists limitations and Section 9 concludes.

2. Model

2.1 Target

All quantities are on the logit scale of Flávio's two-party share of the runoff valid vote,

$$\varphi = \operatorname{logit}(s) = \log \frac{s}{1-s}.$$

Each component k supplies an estimate μ_k with error $e_k = \mu_k - \varphi$. Given the combined estimate $\hat{m}$ and its standard deviation $\hat{\sigma}$ (§2.5), the forecast is

$$\Pr(\text{Flávio wins}) = \Phi\left(\frac{\hat{m}}{\hat{\sigma}}\right), \qquad s \in \operatorname{expit}(\hat{m} \pm 1.96\,\hat{\sigma}) \text{ (95\% interval).}$$

2.2 Components

Bridge (B). A Markov transition from first-round choices to runoff choices (version 7.1). Let X hold the first-round counts of the whole electorate: Lula, Flávio, each eliminated candidate c, residual candidates O, blank/null (BN) and abstention (A). Each row of a row-stochastic matrix P gives where that group goes in the runoff (Lula, Flávio, blank/null, abstention), and $V = X^\top P$:

$$s_B = \frac{V_F}{V_F + V_L}, \qquad \mu_B = \operatorname{logit}(s_B).$$

The rows for Lula (PT), Flávio (PT's opponent), residual candidates, blank/null and abstention are the mean of the 2002–2022 ecological-inference matrices (lphom per state on municipality × zone units; Pavía & Romero 2024). They carry four flows that a "voters stay with their candidate" calculation ignores: retention, defection between finalists, differential abstention (historically 2.2% of PT first-round voters abstain in the runoff against 0.9% of the opponent's) and the runoff choices of first-round blank/null voters (26% Lula-side, 16% opponent-side) and abstainers (0.4% and 1.5%). The row for each eliminated candidate c comes from a pollster transfer grid (f_c, l_c, u_c), read from crosstabs of runoff intention by first-round vote, with two corrections from history. The share that does not vote validly is the historical one for eliminated candidates' voters, $u = 15.5\%$ (10–23% across elections), not the grids' "undecided/blank" share (29–36%). The undecided who do vote split like the committed voters of the same electorate, $q_c = f_c/(f_c + l_c)$:

$$P_{c\cdot} = ((1-u)(1-q_c),\ (1-u)\,q_c,\ u\,\beta,\ u(1-\beta)),$$

with β the historical blank/null share of their non-valid votes. In 2026, $c \in \{$Cury, Renan Santos, Caiado, Zema$\}$, and the grid is the average of the two most recent ones (Genial/Quaest, October; Paraná Pesquisas). Two further grids (Real Time Big Data, July; Quaest, 24–27 September) are reported as a range, and an even split of the undecided as a sensitivity (§6.2). Until version 7.0 the bridge kept every Lula and Flávio voter, ignored blank/null votes and abstention, and dropped the grids' undecided from the denominator.

Pre-first-round polls (H). The sample-size-weighted mean, on the logit scale, of head-to-head Lula × Flávio runoff polls fielded in the 30 days before the first round, shifted by the bias b (§2.3).

Post-first-round polls (F). The same pooled mean over head-to-head polls fielded after the

first round, shifted by b. Its error depends on the horizon (§2.4).

2.3 Bias of Brazilian runoff polls

In each of the six runoffs of 2002–2022, the final-week polls overstated the PT's two-party share:

2002	2006	2010	2014	2018	2022
+3.1	+0.8	+0.7	+1.5	+0.6	+1.2

(pp; 2002–2014 from Jennings & Wlezien 2018, final week; 2018 Datafolha and Ibope, 26–27 October; 2022 mean of Datafolha, Quaest, Ipec, AtlasIntel, MDA and Paraná, 26–29 October.) The mean is $b = 1.31$ pp (sd 0.93, six of six with the same sign). Poll components use $\text{logit}(\bar{s}+b)$, i.e. they are moved towards Flávio. A partially pooled estimate (Shirani-Mehr et al. 2018) with a cross-country prior gives about 1.2 pp; the difference is immaterial.

Undecided voters in runoff polls. Poll components are valid-vote shares, so undecided and blank/null respondents are implicitly split like the decided ones, and whatever they do differently ends up in b. We tested whether b grows with the undecided share, as it would if undecided voters systematically broke against the PT. Across elections, the final polls' undecided/blank share (2002 4.4%, 2006 6.0%, 2010 10.1%, 2022 7.1%) is *negatively* correlated with b (-0.74, $n = 4$). Within elections (41 post-first-round polls in 2014 and 2018, election fixed effects), the slope of the PT two-party error on the undecided share is 0.002 (s.e. 0.15). There is no evidence for scaling b by the undecided share. The 2026 runoff polls have 9.5% undecided/blank on average (interquartile range 7–10%), within the historical range, so the constant b applies (`scripts/analysis_undecided_2T.py`).

The first-round miss is *not* carried into the runoff. Across 21 two-round elections in the Jennings & Wlezien archive, the correlation between the first- and second-round errors of the final polls is 0.06 (slope 0.03). Brazil's own six pairs give 0.49, driven by 2018 and not significant.

2.4 Calibrated errors

Each σ_k is a root-mean-square error against the runoff result, computed on elections other than the one being forecast.

- **Bridge.** For each year 2002–2022, the Markov bridge is computed with that year's first-round counts and the mean transition matrix of the *other* five elections (the eliminated candidates' row also from the other elections, since no historical grids exist). Its RMSE against the runoff is $\sigma_B = 0.168$ logit (≈ 4.1 pp near 50%). A simpler bridge, the PT's first-round share plus the other five elections' mean effective transfer rate times the other candidates' share, has RMSE 0.166: historically, explicit abstention and blank/null flows do not improve the bridge, because the effective transfer rate already absorbs their average. They are kept for 2026 because the grid-based bridge has no such rate to absorb them (§4).
- **Pre-first-round polls.** RMSE of runoff polls fielded before the first round, 11 non-Brazilian runoffs: $\sigma_H = 0.365$.
- **Post-first-round polls.** RMSE of polls fielded after the first round, by days to the runoff (all countries in the archive, including Brazil 2002–2014), interpolated linearly and extrapolated with the last slope beyond 21 days:

Days to runoff	7	14	21
Elections	16	18	7
σ_F (logit)	0.092	0.120	0.163

These replace the hand-set constants of v6 ($\sigma_{\text{drift}} = 0.10$, $\sigma_{\text{house}} = 0.20$, $\sigma_{\text{surprise}} = 0.25$), whose combination ($\approx 0.43$ logit) was only compatible with polls fielded before the first round.

Fewer pollsters than the calibration (pre-registered 6 October 2026). $\sigma_F(h)$ was measured on window averages of about $k_{\text{ref}} = 2.8$ polls (mean poll-days per 7-day window in the archive). An average of k pollsters has the house error that the archive's averages diluted:

$$\sigma^2_{F,\text{eff}} = \sigma_F(h)^2 + \sigma_1^2 \max(0,\ 1/k - 1/k_{\text{ref}}),$$

with k the number of distinct pollsters among post-first-round polls and $\sigma_1 = 0.070$ logit the dispersion of a single poll around its week's mean (105 Lula × Flávio runoff polls in 16 weeks, June–September 2026; sampling plus house error). At 16 days, σ_F goes from 0.132 to 0.135 with two pollsters and 0.143 with one, and is unchanged with three or more.

2.5 Combination with a common drift

Every σ_k is measured against the *result*, so it includes the real movement of opinion between the date of the information and election day. That movement is the same for all components: whatever happens after the more recent of two pieces of information is missed by both. Write each error as a Brownian drift common to all components plus an own error,

$$e_k = D(h_k) + u_k, \qquad \operatorname{Var} D(h) = \sigma_d^2 h, \qquad \operatorname{Var} u_k = \sigma_k^2 - \sigma_d^2 h_k,$$

with u_k independent of each other and of D. Then

$$\Sigma_{kk} = \sigma_k^2, \qquad \Sigma_{ij} = \sigma_d^2 \min(h_i, h_j) \quad (i \neq j),$$

$$\hat{m} = \frac{\mathbf{1}^\top \Sigma^{-1} \mu}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \qquad \hat{\sigma}^2 = \frac{1}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

Σ is a non-negative diagonal plus a Brownian covariance, hence positive definite. With $\sigma_d = 0$ the rule reduces to precision weighting, $\hat{m} = \sum_k \mu_k \sigma_k^{-2} / \sum_k \sigma_k^{-2}$, which is what v7 used until 6 October 2026. Horizons: h_B = days between the rounds (21 in 2026); $h_H = h_B + 15$ (midpoint of the 30-day window); h_F = days remaining to the runoff.

Calibration of σ_d. Regressing the squared logit error of each post-first-round poll average on its horizon, $\mathbb{E}[e(h)^2] = \sigma_0^2 + \sigma_d^2 h$ for $h \leq 28$ days (124 poll-days, 21 elections), gives $\hat{\sigma}_d = 0.0356$ logit/day (90% bootstrap interval over elections 0.019–0.046) and $\hat{\sigma}_0 = 0.087$. This estimate is not internally consistent with the calibrated component errors: $\hat{\sigma}_d^2 \cdot 14$ exceeds $\sigma_F(14)^2$, which would leave the 14-day poll average with no error of its own. The production value is therefore capped so that every calibrated σ_k keeps at least 10% of its variance as own error, $\sigma_d^2 h_k \leq 0.9\, \sigma_k^2$. The binding constraint is the 14-day entry, giving $\sigma_d = 0.0303$. In practice the pure Brownian drift is misspecified and the floor is a modelling choice; §6.4 reports the sensitivity.

2.6 Divergence check (pre-registered)

When post-first-round polls arrive, their disagreement with the bridge is evaluated under the model's own covariance:

$$z = \frac{\mu_F - \mu_B}{\sqrt{\sigma_B^2 + \sigma_{F,\text{eff}}^2 - 2\sigma_d^2 \min(h_B, h_F)}}.$$

The rule was fixed on 6 October 2026, before any post-first-round poll: if $|z| > 2$ the dashboard flags the divergence. The headline number still follows §2.5 (no weight or σ is changed after seeing the data), and a sensitivity with Student-t tails ($\nu = 4$, same variance) is reported next to it. σ_F is not recalibrated on the basis of the disagreement: one or two polls cannot estimate a variance, and widening σ_F because the polls disagree with the bridge would hand weight back to the bridge precisely when new data point elsewhere. With the bridge at 52.3% and polls published around 17 days out, $|z| > 2$ corresponds to a raw two-party average for Flávio below 44.5% or above 57.4% with two pollsters (44.0% and 57.8% with one).

2.7 Inclusion rule

A component enters only if adding it improves the mean predictive log density of the share (the logarithmic score, a strictly proper rule; Gneiting & Raftery 2007) in the leave-one-out backtest. Brier scores are reported but not used for selection: with six confident forecasts that all called the winner, they barely discriminate.

3. Data

- **2026 first round.** TSE totalisation, 100% of polling stations (`resultados.tse.jus.br`, election 6257, office 0001; generated 5 October 2026, 02:59). Valid votes 119,300,788; Flávio 56,104,503; Lula 53,879,538.
- **Historical results.** TSE presidential returns by state, first and second rounds, 2002–2022.
- **International poll archive.** Jennings & Wlezien (2018), Harvard Dataverse doi:10.7910/DVN/8421DX: polls and results for presidential elections with two rounds in Brazil, France, Finland, Peru, Chile, Poland, Romania and Slovenia (1969–2017).
- **2026 polls.** Lula × Flávio runoff head-to-heads from the Wikipedia-PT and Wikipedia-EN poll tables (170 pairs up to 4 October; aggregator averages excluded; documented manual corrections in `data/polls/manual_corrections.csv`).
- **Transfer grids.** Crosstabs of runoff intention by first-round vote: Genial/Quaest (October), Paraná Pesquisas, Real Time Big Data (July), Quaest (24–27 September).
- **Final 2018 and 2022 runoff polls** not in the archive: listed in §2.3.
- **Ecological-inference units.** TSE `votacao_candidato_munzona` and `detalhe_votacao_munzona` files, 2002–2022 (municipality × zone, about 6–12.5 thousand units per election), used in §5 and §7.

4. Validation

Each election of 2002–2022 is forecast with σ_k, t and b estimated on the other five (and, for poll errors, on non-Brazilian runoffs). Two vantage points are tested: the day after the first round

and the final week. Metrics: winner hit rate, Brier score, share RMSE (pp), and mean Gaussian log density of the true logit share.

Vantage	System	n	Hits	RMSE (pp)	Brier	Log density
Day after 1st round	structural (prior + approval)	6	6/6	8.8	0.106	−2.89
	structural + bridge	6	6/6	6.6	0.010	−1.82
	bridge, effective transfer rate	6	6/6	4.1	0.014	−0.02
	Markov bridge (v7.1)	6	6/6	4.1	0.014	−0.29
	bridge + pre-1st-round polls	2	2/2	2.1	0.021	0.77
	same, common drift	2	2/2	2.3	0.029	0.68
Final week	v6 (as published, in-sample)	6	6/6	3.6	0.097	−0.00
	bridge + polls	6	6/6	1.9	0.014	1.03
	bridge + polls, common drift	6	6/6	1.8	0.020	1.11
	bias-corrected polls	6	6/6	1.0	0.033	1.32

Findings:

1. **The structural component is dropped.** Added to the bridge, it raises the share RMSE from 4.1 to 6.6 pp and lowers the log density from −0.02 to −1.82. Its low Brier score comes from confident calls that happened to be right, not from accurate shares.
2. **Pre-first-round head-to-heads enter.** They improve the log density (0.64 → 0.77 on the two testable years). No such polls are published for 2002–2014, so the decision rests on 2018 and 2022.
3. **In the final week, the polls dominate.** Calibrated, bias-corrected polls alone are the best system; adding the bridge with independent errors worsens it. The common-drift rule recovers part of the gap (1.03 → 1.11) because it gives the polls more weight; on the two pre-first-round years it lowers the log density (0.77 → 0.68).
4. **The bridge misses badly once.** In 2006, a 6.9% net defection from Alckmin's first-round voters to Lula left the bridge 7.6 pp off (log density −2.99; Markov bridge 8.1 pp, −4.66). Every other year is within 4.4 pp.
5. **Explicit abstention and blank/null flows: no historical gain.** The Markov bridge is closer than the effective-rate bridge in 4 of 6 elections (2002, 2010, 2014, 2018) and has the same RMSE (4.1 pp), but a lower mean log density (−0.29 against −0.02), entirely because of 2006; without 2006 it is slightly better (0.59 against 0.57). By the inclusion rule (§2.7) it would not replace the effective-rate bridge in the past. It is used in 2026 for a structural reason: the effective rate absorbed turnout and blank/null flows on average, whereas the 2026 bridge is built from grids that contain none of them.

With $n = 6$, none of these differences is statistically significant. The version-6 Brier of 0.097 was in-sample and is not comparable.

5. Tested and rejected

- **Structural prior and approval** (§4): out-of-sample harm.

- **Carry-over of the first-round poll miss** (§2.3): no correlation across 21 elections. The 25/50/100% carry-over sensitivity published in v6.10 had no empirical basis and was removed.
- **Gubernatorial blocs and coalition size.** With the presidential first round known, the state-level left–right gubernatorial bloc reduces state-level RMSE of the PT runoff share by at most 0.1 pp, with coefficients of either sign. Ticket-splitting is large (2022: left-aligned governor candidates 24%, right-aligned 50%, Lula 48% in the presidential first round). Coalition size is collinear with incumbency.
- **Geographic normalisation of transfers.** Ecological inference (lphom linear programming per state; Pavía & Romero 2024) on municipality × zone units for 2014, 2018 and 2022 estimates how the transfer rate of eliminated candidates' voters to the PT's opponent depends on the state's lean: elasticity $\beta = 0.45$ (95% bootstrap interval 0.16–0.65, year fixed effects). Applied to the state distribution of the 2026 minor-candidate vote, it moves the bridge by +0.04 pp: the four candidates' electorates are only 1–4.5 pp more Flávio-leaning than the country, and a national crosstab already integrates the sample's geography.
- **Prediction markets, donations, roll-call cohesion, demographics** (v6 diagnostics): no historical improvement.
- **Institute house effects** (v6.3): even leave-one-out corrections degraded the backtest (Appendix B).

6. Application to 2026

6.1 First-round result

Candidate	Valid votes (%)	Eve-of-election poll average (%)
Flávio Bolsonaro	47.03	42.7
Lula	45.16	44.5
Augusto Cury	2.89	3.7
Renan Santos	2.24	3.7
Ronaldo Caiado	2.18	3.4
Romeu Zema	0.27	1.0

The poll average (latest poll from each of 15 pollsters within 10 days) understated Flávio by 4.3 pp and his two-party share by 2.0 pp, in the same direction as in 2018 and 2022. Following §2.3, this is not carried into the runoff.

6.2 Components on 6 October 2026

Component	Flávio share	σ_k	h_k (days)	Weight
Markov bridge (TSE + transition rows + Quaest/Paraná grids)	52.3%	0.168	21	0.93
Pre-first-round polls (53 pairs, bias-corrected)	51.9%	0.365	36	0.07

Component	Flávio share	σ_k	h_k (days)	Weight
Post-first-round polls	none yet	—	—	—

The averaged grid sends 49.5–55% of each minor electorate to Flávio and 10–19.5% to Lula, with 29–36% uncommitted; with the historical non-valid share (15.5%) and the undecided split like the committed, 71% of their valid runoff votes go to Flávio. The first-round electorate was 158.7 million, with 33.5 million abstentions and 6.0 million blank or null votes. The flows the previous bridge ignored nearly cancel out:

Flow (Markov rows)	Effect on Flávio's margin (million votes)
Defection and abstention of Lula's first-round voters (3.1% leave him)	+1.77
Defection and abstention of Flávio's first-round voters (2.5% leave him, 1.2% to Lula)	−2.07
First-round blank/null voters	−0.58
First-round abstainers	+0.35

The bridge moves from 52.24% (voters stay, undecided dropped) to 52.27%. Across the four grids it ranges from 51.4% to 52.5%; splitting the grids' undecided evenly instead of like the committed gives 51.96% and Pr(Flávio) = 0.681. Combined (§2.5, $\sigma_d = 0.0303$):

$$\hat{s} = 52.2\%, \quad \hat{\sigma} = 0.166, \quad \Pr(\text{Flávio}) = 0.705, \quad \Pr(\text{Lula}) = 0.295,$$

with a 95% interval for Flávio's share of 44.1–60.3%. Under precision weighting the same inputs give $\hat{\sigma} = 0.153$ and Pr(Flávio) = 0.718.

6.3 What the first post-first-round polls will do

The first post-first-round polls (Datafolha, 8 October; AtlasIntel, 9 October) will be about 16 days out, where $\sigma_{F,\text{eff}} \approx 0.135$ with two pollsters. With the common drift and two pollsters they take 79% of the weight. Holding the bridge and pre-first-round polls fixed:

Raw poll average, Flávio two-party	Pr(Lula), common drift	Pr(Lula), independent errors
46%	0.69	0.59
47%	0.60	0.50
48%	0.51	0.41
49%	0.41	0.33
50%	0.32	0.25
51%	0.24	0.18
52%	0.17	0.13
53%	0.12	0.09

Raw poll average, Flávio two-party	Pr(Lula), common drift	Pr(Lula), independent errors
54%	0.08	0.06

Two pollsters are assumed ($\sigma_{F,\text{eff}}$, §2.4). Pr(Lula) = 0.5 requires a raw average of about 48% for Flávio: the bias correction and the bridge together are worth about two points.

6.4 Sensitivity

Varying the own-error floor between 5% and 50% and σ_d across its 90% interval moves today's Pr(Lula) between 0.289 and 0.295, and the 50/50 scenario of §6.3 between 0.282 and 0.329. Splitting the grids' undecided evenly raises today's Pr(Lula) to 0.319. `forecast_v7(..., common_drift=False)` reproduces precision weighting.

7. The comeback probability

No first-round runner-up has won a Brazilian runoff. A separate study (`docs/estudo_cauda_virada_2T.md`) estimates the probability that Lula, 1.87 pp behind, wins on 25 October, with several tools, and keeps only those that survive validation.

Method	Pr(comeback)	Status
Brazilian base rate, 0/7 (Laplace)	0.11 (exact 95%: 0–0.41)	rejected: ignores the margin
International runoffs, 182 cases, logit/probit on margin and free vote	0.35–0.39	upper bound
Markov transition matrix, independent Dirichlet rows	0.09	rejected: under-dispersed
Markov matrix with a common shock calibrated on 2002–2022	**0.26–0.32**	defensible
Random walk over 19 days from the Markov bridge	0.26	consistent
Chaos-theoretic models	—	rejected: not identifiable
v7 (this paper)	0.295	within the defensible range

Base rate and the law of large numbers. With $k = 0$ of $n = 7$, the Clopper–Pearson 95% interval is $[0, 0.41]$. Hoeffding's inequality needs $n \geq \ln(2/\delta)/(2\varepsilon^2) = 738$ runoffs for ± 5 pp at 95%. The seven past margins (5.2–23.2 pp) all exceed 2026's, so under any model where Pr(comeback | m) decreases in the margin m, the base rate is in expectation a *lower bound* for 2026, not an estimator.

International runoffs. In 182 two-round presidential runoffs (44 countries, 1965–2026), 61 were comebacks. A probit on the margin in units of free votes, $m/(o + \Delta T)$, has the best leave-one-out log-loss (0.582 against 0.643 for a constant). Margins below 3 pp flipped 19 times in 40.

These models know how many votes are free but not where they lean: in 2026 about 71% of the minor candidates' valid runoff votes are expected to go to Flávio. They are an upper bound.

Markov chain. The runoff vote is $V = X^\top P$, where X holds the first-round counts (Lula, Flávio, others, blank/null, abstention) and P is a 5×4 row-stochastic transition matrix with Dirichlet rows centred on the 2002–2022 ecological-inference matrices and the 2026 grids. With independent rows it is over-confident: in the leave-one-out backtest its 90% intervals cover 4 of 6 elections and 2006 falls at PIT ≈ 0. Adding a common logit shock $\eta \sim N(0, \tau^2)$, with τ fitted by maximum likelihood on the backtest residuals ($\tau = 0.12$–0.17), gives Pr(comeback) = 0.26–0.32; $\tau \approx 0.165$ independently reproduces σ_B. The comeback boundary is a net defection of about 4.3% of Flávio's first-round voters (about 2.4 million) to Lula; it cannot come from the minor candidates (their transfer to Flávio would have to fall from about 0.71 to 0.39) nor from turnout. The only precedent is 2006 (6.9%).

First passage. By the reflection principle, the latent two-party share crosses 50% at some point before election day in about half of the scenarios (0.50). That is a different event from a comeback and should not be read as one when a poll shows a tie.

Chaos. Voter-model and mean-field Sznajd dynamics are not chaotic (one-dimensional flows), and a logistic map in its chaotic regime is inconsistent with the observed persistence of poll series (lag-1 autocorrelation 0.87). On the available series (69 weekly points in 2026, 26 daily points after the 2022 first round), neither Rosenstein Lyapunov exponents nor nonlinear prediction error differ from linear surrogates. A positive control (logistic map, $r = 4$) shows the test detects chaos in at most 35% of cases at the series' noise level even with 600 points; the Eckmann–Ruelle bound requires $N \gtrsim 10^{D/2}$. Chaos theory adds nothing beyond a variance term already represented by σ_d.

8. Limitations

1. **Six elections.** All Brazilian calibrations (σ_B, b, the inclusion of H) rest on $n = 6$, and the pre-first-round test on $n = 2$. The relative standard error of σ_B is about 30%.
2. **Grids are not the backtested bridge.** σ_B is calibrated on bridges built from a historical mean transfer rate; in 2026 the bridge uses pollster grids, which cannot be backtested. Their subsamples are small (20–150 interviews per candidate) and predate the first round.
3. **Turnout and undecided flows come from averages.** The transition rows are six-election means from ecological inference, which produces corner solutions and leans on 2006 for defection. How the grids' undecided will split cannot be checked against history; the even-split alternative moves Pr(Lula) by +0.024. The runoff turnout of 2026 is unknown, and a mobilisation pattern unlike 2002–2022 would not be captured.
4. **The drift model is misspecified** (§2.5). The effective σ_d is set by an own-error floor, not estimated.
5. **International poll errors** come from eight countries, mostly European, and may not transfer to Brazil's polling industry. The Brazilian bias correction partly addresses this.
6. **Normal errors.** Tails are Gaussian on the logit scale. With $n = 6$ a Student-t cannot be identified; at the current distance from the boundary ($\approx 0.5\hat{\sigma}$) heavier tails would *lower* Pr(comeback).
7. **Ecological inference** uses lphom, which produces corner solutions; the Dirichlet-

multinomial estimator of Rosen et al. (2001) is the natural replacement.

9. Conclusion

After the first round, Brazil's runoff is forecast best by combining the first-round result with the polls, each weighted by an error measured on past elections, and not by structural fundamentals. The first-round bridge is the anchor until runoff polls arrive; within days of the runoff the polls, corrected for a small and persistent pro-PT bias, dominate. Treating the components' errors as independent overstated the certainty of the combination because they share the drift of opinion to election day; the generalised-least-squares rule fixes that at the cost of an imperfectly identified drift variance.

On 6 October 2026 the model gives Flávio Bolsonaro a 70.5% chance of winning the runoff and Lula 29.5%, a comeback that would be a first for Brazil but that the evidence places at roughly one in three or four. The number will move most with the first post-first-round polls, and the quantity to watch in them is the share of Flávio's first-round voters who say they will vote for Lula.

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References for the superseded v6 framework are listed in Appendix B.

Policy / disclosure guidance on generative AI in scholarship

- Nature Editorial (2023). *Tools such as ChatGPT threaten transparent science; here are our ground rules for their use.* Nature, 613(7945): 612.
- Science Editorial (2023, updated 2024). *ChatGPT is fun, but not an author.* Science, 379(6630): 313.
- Journal of the American Statistical Association — Editorial Policies (2024). *Use of generative AI in submissions.* American Statistical Association.

Data Sources

- **TSE — 2026 results.** https://resultados.tse.jus.br/oficial/ele2026/6257/dados/br/br-c0001-e006257-u.json (first round, president, national).
- **TSE — Open Data Portal.** https://dadosabertos.tse.jus.br/; bulk files `votacao_candidato_munzona_<year>.zip` and `detalhe_votacao_munzona_<year>.zip` at https://cdn.tse.jus.br/estatistica/sead/odsele/.
- **Jennings & Wlezien (2018) archive.** https://doi.org/10.7910/DVN/8421DX.
- **Wikipedia — poll tables.** *Pesquisas eleitorais para a eleição presidencial no Brasil em 2026* (PT) and *Opinion polling for the 2026 Brazilian presidential election* (EN).
- **International runoff results** for the comeback study: Wikipedia EN election articles, one URL per row in `data/estudo_cauda/runoffs_presidenciais.csv`.

Software and Toolchain

- **Python 3.13** with `numpy`, `scipy`, `pandas`.
- **Pandoc** and **Tectonic** for this paper; the dashboard renders it as HTML with native MathML.
- **React 18** and **Chart.js 4** (inlined) for the companion dashboard.
- **Anthropic Claude** for drafting assistance, as disclosed above.

Appendix A — Reproducibility

```
<project-root>/
|-- src/
|   |-- model_v7.py          # components, calibration loading, GLS combination
|   |-- first_round.py       # first-round average and bridge
|   `-- data_2026.py         # transfer grids, TSE result loader
|-- scripts/
|   |-- fetch_first_round_result.py  # TSE 2026 first round
|   |-- fetch_polls_live.py          # Wikipedia poll tables
|   |-- aggregate_live_to_scenarios.py
|   |-- backtest_v7.py               # leave-one-out validation and calibration
|   |-- project_2026.py              # 2026 forecast
|   `-- estudo_cauda/                # comeback study (Section 7)
|-- data/  results/  docs/
`-- build_dashboard.py               # dashboard
```

```
PYTHONPATH=. python scripts/fetch_first_round_result.py
PYTHONPATH=. python scripts/fetch_polls_live.py
PYTHONPATH=. python scripts/aggregate_live_to_scenarios.py
PYTHONPATH=. python scripts/backtest_v7.py      # writes results/v7_calibration.json
PYTHONPATH=. python scripts/project_2026.py     # writes results/projection_2026.json
PYTHONPATH=. python build_dashboard.py
```

Appendix B — The v6 framework (superseded)

Versions 6.0–6.10 (May–October 2026) are reproduced below as published, for the record. They forecast the runoff before the first round from head-to-head polls and a structural prior driven by government approval, marginalised over the identity of the right-wing candidate. Version 7.0 (Sections 1–9) replaces them. Section references inside this appendix point to its own numbering (B.x). Numbers are those of the v6 runs and are not updated.

B.0 Version history v5.0–v6.3

Changes in v6.3 (May 2026). Following an internal peer review documented in `PEER_REVIEW.md`, four bugs were identified and corrected, and the institute-bias correction was abandoned:

R7.1 — Data leakage in house-effects corrections (S1 of review). The v5/v6 institute-bias model estimated b_h from residuals $r_{j,h,t} = y_{h,t,\text{real}} - p_{j,h,t}$ over *all* elections $t \in \{2002, \ldots, 2022\}$ and then applied those corrected polls back into the *same* elections during walk-forward backtesting, producing an in-sample leak. A leave-one-out reimplementation (excluding the target election from b_h training) was developed and tested. Empirically, *both* the leaky in-sample correction (Brier 0.132) and the honest LOO correction (Brier 0.115) degrade the backtest relative to raw polls (Brier 0.097 with the v6.3 fixes below). Diagnosis: the model $b_h \sim$ institute is mis-specified because the identity of candidate A rotates between PT, PSDB, and PL across elections; aggregating residuals across these introduces systematic mis-correction. **Institute-bias corrections are abandoned in v6.3**; polls are used raw.

R7.2 — Precision double-counting in poll fusion (S2 of review). The flat-fit fusion in `model_2T.fit_2T` computed effective precision as $w_i \cdot k_i/\sigma_i^2$ with $k_i = N_{\text{polls}}$ for opinion polls, but σ_i was already the *pooled* standard deviation $\sigma_{\text{pool}} = (\sum_j 1/\sigma_j^2)^{-1/2}$, which itself incorporates N. The product over-weighted polls by a factor of N, so 252 polls in 2022 dominated approval by a ratio of $\approx 27{,}000 : 1$. Sensitivity tests of approval ± 20 pp on 2022 produced *exactly* zero variation in $\Pr(A)$. v6.3 replaces the formula with w_i/σ_i^2 (correct for a Gaussian observation already pooled). A precision cap is additionally imposed via $N_{\text{eff,max}} = 30$ "equivalent" polls so that 252 polls never overwhelm the approval signal entirely. The hierarchical path (`fit_2T_hierarchical`, v6.2) did not have this bug.

R7.3 — Incumbency strength as a continuous covariate (M5 of review). The Boolean `incumbent_is_A` flag was replaced with a continuous

`incumbent_strength_A` `[0,1]` that modulates the sign of the approval coefficient: sign $= 2 \cdot \text{strength} - 1$. In the historical sample this is 0.0 for the canonical incumbent-versus-challenger races (2002 Serra/PSDB, 2006/2010 Lula-Dilma/PT, 2014 Dilma/PT, 2022 Bolsonaro/PL) and 0.5 for 2018 (Haddad/PT inherits a symbolic incumbency after the 2016 PT impeachment, with sign = 0 neutralizing the approval signal — Temer/MDB was the formal incumbent but did not reach the runoff). This eliminates a sign-misapplication that had been masking 2018's true signal load.

R7.4 — Other (S4 / M1 of review). The transfer-matrix intercept θ_0 in `transfer_matrix` had been silently zeroed via a stray `theta0 * 0` (fixed); the irreducible-variance scaling in the deprecated multiclass 1T path was applying the time scaling to *all* three components rather than to drift alone (fixed and the entire 1T multiclass path was archived).

Updated headline metrics (6 runoffs, 2002–2022): Brier 0.097 (v6.2 had reported 0.132 but the reproducible run with the v6.2 code gives 0.132 — confirming the value with the bug), accuracy 6/6 (genuinely confident, no P=0.500 ties), mean absolute share error 2.7 pp. Drop-one shows both core sources are now load-bearing: $\Delta\text{Brier}_{\text{polls}} = +0.016$ and $\Delta\text{Brier}_{\text{appr}} = +0.035$ (the v6.2 model had a *negative* polls delta — an artifact of the leaky corrections). Stress on ± 20 pp Lula approval in 2022 now produces a ± 2.6 pp swing in $\Pr(A)$ (zero in v6.2).

The 2026 forecast is **unchanged** at $\Pr(\text{right}) = 0.502$ because the hierarchical path used for projection did not carry the precision double-count.

Changes from v5.0 to v6.2 (preserved for completeness). Four methodological revisions were introduced:

R1 (parsimonious core). The model is simplified from a six-source fusion to a **two-source core** (opinion polls and government approval) after head-to-head empirical evidence showed that markets, donations, roll-call cohesion, and demographics contributed ≤ 0.001 to the Brier score and injected ECE-degrading noise in data-sparse elections. The four demoted sources are still computed and reported as **auxiliary out-of-posterior diagnostics**.

R2 (two-way random-effects house effects). The institute-only house-effect model is generalized to a two-way decomposition separating persistent institute bias b_h from election-wide swing δ_t. Empirical decomposition attributes 33.8% of the historical poll-residual variance to election-wide δ_t — a third of what v5 erroneously assigned to institutes. The realigned institute biases cluster much closer to zero, while Veritá's right-leaning posture and the documented 2014/2018 shy-voter effects emerge cleanly.

R3 (recalibrated source weights — v6.1). A formal sensitivity audit of the historical posterior reveals that in poll-rich elections the approval signal — though structurally large — was *operationally swamped* by poll precision (ratios of 100:1 to 100,000:1 in 2010/2018/2022). Recalibrating $(\bar{\gamma}_{\text{polls}}, \bar{\gamma}_{\text{appr}})$ from $(0.55, 0.45)$ to

$(0.30, 0.70)$ recovers the 2010 election (Lula approval +55 pp peak previously masked by noisy polls) and restores 6/6 accuracy with Brier 0.132 (vs. 0.135 in v6.0).

R4 (hierarchical candidate model — v6.2). To resolve the "candidate-blind" pathology in lean v6.1 — under which candidates without dedicated polls collapse to the same probability — we introduce a bivariate Gaussian decomposition $\varphi_k = \varphi_{\text{base}} + \delta_k$. Approval observes φ_{base}; polls and structural expert anchors observe $\varphi_{\text{base}} + \delta_k$ jointly. The closed-form posterior is multivariate Gaussian, with marginal φ_k recovered analytically. This enables candidate-specific probabilities even when polls are absent, via expert anchors with explicit uncertainty $\sigma_{x,k}$.

Net effect on the 2026 forecast. With the four-candidate succession field (Tarcísio added as the post–Banco Master PL natural successor), a −1.5 pp event-shift on Flávio polls, and a composite empirical anchor for Tarcísio ($0.57 \cdot \text{pop}_{\text{SP}} + 0.51 \cdot \text{pop}_{\neg\text{SP}} = 0.523$ national), the v6.2 marginalized estimate is Pr(right wins) = 0.502 (Lula 0.498) — a sharp narrowing from v6.1's 0.541, driven by the now-distinguishable, structurally weak Renan scenario and the empirically anchored Michelle scenario.

B.2 Methodology

B.2.1 Master equation For an election t with a runoff between candidates A (the *challenger*: opposition to the incumbent government) and B (the *incumbent / governist*), let

$$\theta_t \;=\; \Pr(A \text{ wins runoff}), \qquad \varphi_t \;=\; \text{logit}(\theta_t) \;=\; \log\frac{\theta_t}{1-\theta_t}.$$

Inference is conducted in φ-space, where all source likelihoods are approximately Gaussian by delta-method arguments.

Given a structural prior

$$\pi(\varphi_t) \;=\; \mathcal{N}(\mu_0,\ \sigma_0^2),$$

and source-specific likelihoods

$$\ell_i(y_i \,|\, \varphi_t, \theta_i) \;=\; \mathcal{N}(\hat{\varphi}_i,\ \sigma_i^2), \qquad i = 1, \ldots, 6,$$

with effective sample size k_i and source weight γ_i, the posterior is closed-form Gaussian. Defining the *precision* (inverse variance) and *precision-weighted mean*, we obtain

$$\boxed{\begin{aligned} \frac{1}{\sigma^2_{\text{post}}} \;&=\; \frac{1}{\sigma_0^2} \;+\; \sum_{i=1}^{6} \frac{\gamma_i\, k_i}{\sigma_i^2}, \\ \mu_{\text{post}} \;&=\; \sigma^2_{\text{post}} \left[\frac{\mu_0}{\sigma_0^2} \;+\; \sum_{i=1}^{6} \frac{\gamma_i\, k_i\, \hat{\varphi}_i}{\sigma_i^2} \right]. \end{aligned}}$$

The predictive probability that A wins the runoff is then

$$\Pr(A \text{ wins}) \;=\; \Phi\left(\frac{\mu_{\text{post}}}{\sqrt{\sigma^2_{\text{post}} \;+\; \sigma^2_{\text{irr}}}}\right),$$

where $\Phi(\cdot)$ is the standard-normal CDF and σ^2_{irr} captures irreducible uncertainty (drift, residual house effects after correction, and structural surprise; see §B.2.4).

B.2.2 Source likelihoods The model splits the six available data sources into a **core** ($\mathcal{S}_{\text{core}}$ = {polls, approval}, used to construct the posterior) and **auxiliary diagnostics** ($\mathcal{S}_{\text{diag}}$ = {market, donations, roll-call, demographics}, computed and reported but assigned $\gamma_i = 0$). The empirical justification for this split is the drop-one source-sensitivity result of §B.5.4 (Table 4), which shows that the diagnostic sources move the Brier score by less than 0.001.

Each diagnostic source is reported alongside the posterior with its implied logit-space tilt $\hat{\varphi}_i$, allowing readers to identify cases where the diagnostics disagree with the core forecast — a signal that the *out-of-equation* qualitative layer (§B.7) should be invoked.

B.2.2.1 Opinion polls (Gaussian-ALR approximation) — *core* For each runoff poll j from pollster h, conducted d_j days before the election with sample size n_j and reported share $\hat{p}_{A,j}$ for the challenger:

$$\hat{\varphi}_j \;=\; \text{logit}(\hat{p}_{A,j}), \qquad \hat{\varphi}_j \,|\, \varphi_t \;\sim\; \mathcal{N}\left(\varphi_t + b_h,\;\; \sigma^2_{\text{obs}}\,\frac{n_{\text{ref}}}{n_j} \;+\; \sigma^2_\eta\, d_j\right),$$

where b_h is the institute-specific house effect (Section 4) and the variance decomposes into a sampling term (which scales as $1/n_j$) and a latent-drift term (which scales linearly with the time gap d_j). We set $\sigma_{\text{obs}} = 0.30$ in logit space (≈ 5 pp on the share scale), $n_{\text{ref}} = 1{,}500$, and $\sigma_\eta = 0.015$ logit/day.

This formulation departs from the multinomial likelihood of multi-candidate models. By treating each poll as one Gaussian observation, the per-source log-likelihood scales linearly with the *number of polls*, not with total respondents. This prevents poll evidence from dominating other information sources by orders of magnitude—a structural failure we encountered in earlier multi-candidate versions of the model.

B.2.2.2 Prediction markets — *auxiliary diagnostic* Let π_k denote the price of contract k on Polymarket or Kalshi for the event "candidate A wins overall". Following Thaler & Ziemba (1988) and Snowberg & Wolfers (2010), we debias for the favorite-longshot effect and order-book spread ϕ via

$$\hat{p}_k \;=\; \frac{\pi_k \;-\; \lambda(\pi_k) \;-\; \phi/2}{1 \;-\; \lambda(\pi_k) \;-\; \mu(\pi_k) \;-\; \phi},$$

with the bias kernels

$$\lambda(\pi) \;=\; c_1\,\pi\,(1-\pi)^2, \qquad \mu(\pi) \;=\; c_2\,\pi^2\,(1-\pi).$$

The corrected price is then mapped to logit space as $\hat{\varphi}_k = \mathrm{logit}(\hat{p}_k)$, with $\sigma_{\mathrm{mkt}} = 0.30$. Markets enter the master equation with weight zero before 2018 (no liquid Polymarket contracts), and partial weight in 2018–2022 as liquidity ramped up.

B.2.2.3 Government approval (Time-for-Change adapted) — *core* Following the *Time-for-Change* tradition (Abramowitz 1988) adapted for the Brazilian context, the contribution of incumbent approval to φ_t is

$$\hat{\varphi}_{\mathrm{inc}} \;=\; \alpha_0 \;+\; \alpha_1\,A_{\mathrm{avg\,3m}} \;+\; \alpha_2\,\frac{\Delta\mathrm{GDP}}{5} \;+\; \alpha_3\,\mathrm{trend}_{\mathrm{6m}},$$

where $A_{\mathrm{avg\,3m}}$ is the average net approval (positive minus negative) over the 3 months preceding the runoff, $\mathrm{trend}_{\mathrm{6m}}$ is the 6-month linear slope, and ΔGDP is the contemporaneous real GDP growth (in percent). We set

$$\alpha_1 = 1.8, \quad \alpha_2 = 0.12, \quad \alpha_3 = 0.8, \quad \sigma_{\mathrm{appr}} = 0.30,$$

which capture the empirical regularity that 20 pp of net approval translates to roughly 4 pp of vote share. The sign of $\hat{\varphi}_{\mathrm{inc}}$ is flipped when A is the incumbent.

B.2.2.4 Electoral donations (TSE) — *auxiliary diagnostic* Let $D_A, D_B \in \mathbb{R}_{>0}$ denote total nominal donations (party fund + electoral fund + private donations) in BRL through $T-30$ days, for candidates A and B respectively. Then

$$\hat{\varphi}_{\mathrm{don}} \;=\; \delta\,\log\frac{D_A}{D_B}, \qquad \sigma_{\mathrm{don}} \;=\; 0.40, \qquad \delta = 0.20.$$

The log-ratio captures the asymmetric pattern in which extreme donation gaps matter more than marginal ones, while keeping the contribution bounded.

B.2.2.5 Roll-call coalition cohesion — *auxiliary diagnostic* For each pair (A, B), we compute the cohesion of the respective Senate party coalitions during the legislature preceding the election. For party p and roll-call vote v, let $n^{\mathrm{yes}}_{p,v}$ and $n^{\mathrm{no}}_{p,v}$ denote the number of senators of party p voting Yes and No on v, with $n_{p,v} = n^{\mathrm{yes}}_{p,v} + n^{\mathrm{no}}_{p,v}$ the total number of present and decided voters. The cohesion of party p is

$$c_p \;=\; \mathbb{E}_v\left[\,\frac{\max\!\left(n^{\mathrm{yes}}_{p,v},\; n^{\mathrm{no}}_{p,v}\right)}{n_{p,v}}\,\right],$$

restricted to votes with $n_{p,v} \geq 10$ to reduce noise. The contribution to φ_t is then

$$\hat{\varphi}_{\mathrm{rc}} \;=\; \theta_c\,(c_A - c_B), \qquad \theta_c = 1.0, \qquad \sigma_{\mathrm{rc}} = 0.50.$$

The intuition is that high cohesion implies effective coalition mobilization in the runoff campaign.

B.2.2.6 Demographic MRP — *auxiliary diagnostic* A multilevel regression with post-stratification (MRP) over IBGE PNAD-C and Census 2022 cells produces an estimate $\hat{p}_A^{\text{MRP}}$, which enters the master equation as

$$\hat{\varphi}_{\text{demo}} \;=\; \text{logit}(\hat{p}_A^{\text{MRP}}), \qquad \sigma_{\text{demo}} \;=\; 0.45.$$

This source is currently implemented as a placeholder; a full MRP implementation requires microdata access beyond the open SIDRA endpoint and is left to future work.

B.2.3 Dynamic weights In the **v6.1 lean specification** (revised in R3, see preface), only the two core sources enter the posterior; their prior weights are

$$\bar{\gamma}_{\text{polls}} = 0.30, \qquad \bar{\gamma}_{\text{appr}} = 0.70,$$

with the four auxiliary diagnostics carrying $\bar{\gamma}_i = 0$ identically. This is a **recalibration** from the v6.0 lean defaults of $(0.55, 0.45)$. The motivation, derived from the sensitivity audit of §B.5.6, is that in poll-rich elections (2010, 2018, 2022) the structural sensitivity of the approval likelihood — $\alpha_1 = 1.8$ in logit-space — was *operationally swamped* by poll precision. The ratio of effective precisions $\text{prec}_{\text{polls}}/\text{prec}_{\text{appr}}$ ranged from ≈ 100 (2006 with 9 polls) to $\approx 113{,}000$ (2022 with 252 polls), rendering approval inert in those cycles. The 2010 election — Lula's net approval at its all-time peak of +55 pp, the textbook Time-for-Change signal — was misforecast at 0.50/0.50 under v6.0 because polls in that cycle were neutral. The recalibrated $(0.30, 0.70)$ split recovers 2010 (now correctly favoring Dilma) and improves Brier by 0.003 on backtest, with the cost concentrated in slightly less aggressive in-sample fits to poll-coherent cycles.

The effective weight in election t is still modulated by availability and reliability,

$$\gamma_i(t, \tau) \;=\; \alpha_i(t, \tau) \cdot \beta_i(t) \cdot \bar{\gamma}_i,$$

where:

- $\alpha_i(t, \tau) \in [0, 1]$ is the **availability factor** (e.g., zero when no runoff polls are available for the candidate pair, as in 2002);
- $\beta_i(t) \in [0, 1]$ is a **reliability factor**, calibrated by historical out-of-sample error;
- $\bar{\gamma}_i \in [0, 1]$ is the **prior weight**, fixed across elections.

When polls are unavailable ($\alpha_{\text{polls}} = 0$), approval carries the entire data signal of the posterior, with the prior (μ_0, σ_0^2) providing the remaining structure — a regime that occurs both for the 2002 historical case and for several 2026 succession scenarios (Michelle, Renan, Tarcísio).

B.2.4 Irreducible uncertainty To prevent posterior over-confidence, we inflate the posterior variance by an irreducible component decomposed into three orthogonal sources of error — latent drift, residual house effects after correction, and structural surprise:

$$\sigma^2_{\text{irr}} \;=\; \sigma^2_{\text{drift}}\,\frac{\tau}{7} \;+\; \sigma^2_{\text{house}} \;+\; \sigma^2_{\text{surprise}},$$

with parameters (in logit-space)

$$\sigma_{\text{drift}} = 0.10, \quad \sigma_{\text{house}} = 0.20, \quad \sigma_{\text{surprise}} = 0.25.$$

The drift term scales linearly with the forecast horizon τ (in days) following random-walk intuition; the house and surprise components are constants of the model.

B.2.5 Bivariate hierarchical extension (v6.2) The v6.1 lean posterior reduces, when polls are absent, to a function of approval alone — and, by construction, of *only* approval. In a forecasting context with multiple candidate scenarios sharing the same incumbent (and hence the same approval signal), all unpolled candidates collapse to the *same* posterior probability. This was the "candidate-blind" pathology documented in v6.1 against the 2026 succession field, where Michelle, Renan, and Tarcísio — each with distinct structural profiles — yielded identical $\Pr(A \text{ wins})$.

The v6.2 specification resolves this with a **bivariate Gaussian hierarchical decomposition** of the latent state. Let

$$\theta \;=\; (\varphi_{\text{base}},\, \delta_k)^\top \in \mathbb{R}^2,$$

where φ_{base} is the *macro mood* of the election (the latent state common to all candidates from the same bloc, driven by the incumbent's approval and the structural prior), and δ_k is the *candidate-specific deviation* from baseline. The total predictive logit is then $\varphi_k = \varphi_{\text{base}} + \delta_k$.

B.2.5.1 Prior Independent Gaussian priors:

$$\varphi_{\text{base}} \sim \mathcal{N}(0,\; \sigma_b^2), \qquad \delta_k \sim \mathcal{N}(0,\; \sigma_\delta^2),$$

with $\sigma_b = 1.5$ (broad prior on macro mood) and $\sigma_\delta = 1.0$ (broad prior on candidate effect — approximately ± 25 pp in share space). Together these encode the *a priori* equally likely outcome under no information.

B.2.5.2 Likelihoods — linear observations $H\theta$ All three information sources are linear functionals of θ:

Source	Observation	H
Approval	$y_a \sim \mathcal{N}(\varphi_{\text{base}},\; \sigma_a^2)$	$(1, 0)$
Polls	$y_p \sim \mathcal{N}(\varphi_{\text{base}} + \delta_k,\; \sigma_p^2)$	$(1, 1)$
Anchor	$y_x \sim \mathcal{N}(\varphi_{\text{base}} + \delta_k,\; \sigma_x^2)$	$(1, 1)$

Critically, **polls observe the total logit φ_k, not δ_k alone**. Polls are real measurements of voting intention, which jointly reflect macro mood and candidate appeal. Treating polls as

observations of δ_k conditional on a point estimate of φ_{base} — a tempting shortcut — produces a mathematically incoherent posterior that breaks the backtest (Brier 0.162, accuracy 5/6 in our reference experiment).

The *expert anchor* y_x is a structural estimate of φ_k — typically a composite of regional sub-polls or transferred historical shares — with explicit uncertainty σ_x reflecting the strength of the empirical evidence (e.g., $\sigma_x = 0.20$ for Tarcísio's composite, $\sigma_x = 0.28$ for Michelle, $\sigma_x = 0.35$ for Renan).

B.2.5.3 Closed-form posterior By standard Gaussian conjugacy with linear observations, the posterior precision is

$$\Sigma_{\text{post}}^{-1} \;=\; \Sigma_{\text{prior}}^{-1} \;+\; \sum_{i \in \{a,p,x\}} \frac{1}{\sigma_i^2}\, H_i^\top H_i,$$

with prior precision $\Sigma_{\text{prior}}^{-1} = \text{diag}(\sigma_b^{-2},\, \sigma_\delta^{-2})$, and the posterior mean is

$$\mu_{\text{post}} \;=\; \Sigma_{\text{post}} \left[\sum_{i \in \{a,p,x\}} \frac{y_i}{\sigma_i^2}\, H_i^\top \right].$$

The marginal in $\varphi_k = [1, 1]\,\theta$ is

$$\begin{aligned} \mathbb{E}[\varphi_k \mid \text{data}] \;&=\; [1,1]\,\mu_{\text{post}}, \\ \text{Var}[\varphi_k \mid \text{data}] \;&=\; [1,1]\,\Sigma_{\text{post}}\,[1,1]^\top \\ &=\; \sigma^2_{\varphi_{\text{base}}} \;+\; \sigma^2_{\delta_k} \;+\; 2\,\text{Cov}(\varphi_{\text{base}},\, \delta_k). \end{aligned}$$

The covariance term — typically negative — encodes the *trade-off* between attributing observed variation to macro mood vs. candidate effect: when polls disagree with approval, the model partitions the discrepancy between the two latent components in proportion to their prior precisions, and the resulting posterior covariance is negative.

The predictive probability is $\Pr(A \text{ wins}) = \Phi\big(\mu_k / \sqrt{\text{Var}(\varphi_k) + \sigma^2_{\text{irr}}}\big)$ as before.

B.2.5.4 Interpretation and trade-offs The hierarchical decomposition has three desirable properties:

1. **Candidate identifiability**: unpolled candidates now receive distinct posterior probabilities, driven by their expert anchors;
2. **Honest uncertainty inflation**: scenarios with weaker anchors (larger σ_x) automatically receive wider posteriors;
3. **Coherence under perturbation**: stress on approval moves φ_{base}, but δ_k adjusts via the posterior covariance to maintain consistency with observed polls. This is the *informationally coherent* behavior — polls anchor candidate-specific deviation regardless of macro shocks.

The cost is a calibration penalty in the historical backtest where polls are abundant: the σ_δ

prior contributes uncertainty even when δ_k is well-identified by data, inflating posterior variance by ~ 0.05 in logit-space. The empirical penalty is $\Delta\text{Brier} = +0.013$ (from 0.132 to 0.145) on the six-election backtest, with accuracy preserved at 6/6 (see Table 2b). We deem this acceptable because the practical benefit (distinguishing 2026 candidates) outweighs the in-sample fit loss.

B.3 Data

Raw data and processing scripts are version-controlled in the project's research repository (access on request).

Source	Format	Coverage	Provider
Official TSE results	CSV, 7 elections × 1T+2T	1998-2022	TSE (mirrored via Wikipedia + manual audit of 2018)
Polls (1T+2T)	CSV, **356 runoff polls**	2006-2022 (no 2002)	Wikipedia poll aggregators with dedicated BeautifulSoup parser
Approval	CSV, 78 quarterly points	1998 Q3–2026 Q2	Compiled from Datafolha/CNT/Ipsos/Quaest published series
Donations	CSV, top 4 candidates × 6 elections	2002-2022	TSE accounts (compiled)
Senate roll-calls	1,767 votes, 142k individual votes	2011-2025	legis.senado.leg.br open data API
Chamber roll-calls	359 plenary votes	2019-2025	dadosabertos.camara.leg.br
IBGE PNAD-C + Census 2022	6 tables × 27 UFs	annual	apisidra.ibge.gov.br
State-level 2026 polls	12 UFs (79% of population)	April-May 2026	Aggregated by Paragon Capital (Maynard 2026)

The poll parser is the most consequential data artifact. Its design follows three principles: (i) section-aware extraction (we identify runoff polls via HTML `h2`/`h3` headings labeled "Segundo turno" rather than generic table mining), (ii) accent-tolerant candidate matching using NFKD normalization, and (iii) a fallback heuristic that detects share columns by content (largest two percentages in the 30–65% range) when the header-based approach fails on sub-header tables.

B.4 House effects via partial pooling

> **v6.3 status — DEPRECATED.** The institute-bias correction described in this section was abandoned in v6.3 after the peer review found that *both* the in-sample (leaky) and leave-one-out (honest) variants degrade the backtest relative to using raw polls. Root cause: the identification $b_h \sim$ institute is mis-specified because the

identity of candidate A rotates between PT, PSDB, and PL across elections. Section retained for completeness; the modules the institute-bias modules are now archived. See `PEER_REVIEW.md` §S1 for the empirical demonstration.

B.4.1 Motivation Walk-forward diagnostics on the v4 model (which ignored house effects) revealed two stylized facts in the 356-poll database:

1. **Almost all pollsters systematically overestimate the challenger** (mean signed residuals 6 to −1 percentage points across institutes).
2. **Institute Veritá is the conspicuous exception**, with *positive* residuals (+1.9pp), consistent with the qualitative finding by Maynard (2026) that this institute carries a known right-leaning bias.

A direct, unweighted bias subtraction would over-correct institutes with few historical polls. We therefore implement empirical-Bayes partial pooling (Jackman 2005; Linzer 2013).

B.4.2 Model — two-way random-effects decomposition Let $s^{\text{real}}_{A,t}$ be the realized share of candidate A in election t, and let $\hat{p}_{A,j,h,t}$ be the share reported by poll j of institute h in election t. Define the residual

$$r_{j,h,t} \;=\; s^{\text{real}}_{A,t} \;-\; \hat{p}_{A,j,h,t}.$$

The **v5 model** assumed all systematic discrepancy came from the institute: $r_{j,h,t} \sim \mathcal{N}(b_h, \sigma_h^2)$. Empirical evidence (Quaest's flipped bias between 2018 and 2022; the well-documented industry-wide underestimation of Bolsonaro in 2018; etc.) suggests this is misspecified.

v6 model. We adopt the standard two-way random-effects decomposition (Searle, Casella, McCulloch 1992; Gelman & Hill 2006):

$$\begin{aligned} r_{j,h,t} \,\big|\, b_h, \delta_t \;&\sim\; \mathcal{N}(b_h + \delta_t,\; \sigma^2_{\text{obs}}), \\ b_h \;&\sim\; \mathcal{N}(0,\; \tau_h^2), \\ \delta_t \;&\sim\; \mathcal{N}(0,\; \tau_t^2), \end{aligned}$$

where b_h captures the *persistent* institute bias (methodology, fieldwork asymmetry, weighting choices) and δ_t captures the *transient* election-wide swing common to all institutes (late-campaign movements, shy-voter effects, turnout surprises). Identification requires the standard sum-to-zero constraints

$$\sum_h N_h\, b_h \;=\; 0, \qquad \sum_t \delta_t \;=\; 0.$$

B.4.3 Estimation by alternating-least-squares With $|H| = 17$ institutes, $|T| = 6$ elections and $N = 356$ polls, we solve by ALS iteration. At each pass:

$$b_h \;\leftarrow\; \text{weighted mean}_j\Big[\, r_{j,h,t} - \delta_t,\; w_h(t) \,\Big],$$
$$\delta_t \;\leftarrow\; \text{mean}_j\Big[\, r_{j,h(j),t} - b_{h(j)} \,\Big],$$

with weights $w_h(t) = \exp(-(2026 - t)/\tau_{\text{decay}})$, $\tau_{\text{decay}} = 8$ years, to favour recent calibration. Identification constraints are re-imposed after each update. Convergence to 10^{-6} is achieved in 5–8 iterations.

B.4.4 Estimated variance components The decomposition yields

$$\hat{\sigma}^2_{\text{obs}} \;=\; 0.00384, \qquad \hat{\tau}^2_h \;=\; 0.00113, \qquad \hat{\tau}^2_t \;=\; 0.00254.$$

Table HE.1. Variance attribution.

Component	Variance	Std (pp on share)	Share of total
σ^2_{obs} (within cell)	0.00384	6.20	51.2%
τ^2_h (institute persistent)	0.00113	3.36	15.1%
τ^2_t (election-wide swing)	0.00254	5.04	**33.8**%

One-third of the systematic poll-residual variance was election-wide δ_t, not institute-specific b_h. A model assigning everything to b_h — as v5 did — necessarily over-corrects institutes in elections where they collectively missed by a common δ_t.

The shrunk institute estimates (James-Stein under the new prior τ^2_h):

$$\hat{b}^{\text{shrunk}}_h \;=\; \frac{N^{\text{eff}}_h}{N^{\text{eff}}_h + k_h}\,\bar{r}^{\text{adj}}_h, \qquad k_h \;=\; \frac{\sigma^2_{\text{obs}}}{\tau^2_h},$$

with $\bar{r}^{\text{adj}}_h$ the weighted average institute residual *after subtracting the estimated δ_t for each poll's election.*

B.4.5 Estimated institute biases (v6 vs v5) **Table HE.2.** Institute-level shrunk biases (pp on share scale). The v5 column attributed all variability to the institute; the v6 column properly isolates the persistent component.

Institute	N_h	v5 $\hat{b}^{\text{shrunk}}_h$	**v6+R2** $\hat{b}^{\text{shrunk}}_h$
Datafolha	30	−5.6	**−3.1**
Quaest	27	−5.7	**−2.8**
Ipespe	36	−6.1	**−2.6**
PoderData	41	−5.0	−1.6
CNT/MDA	13	−4.3	−1.6
Atlas	15	−3.6	+0.5

Institute	N_h	v5 $\hat{b}_h^{\text{shrunk}}$	**v6+R2** $\hat{b}_h^{\text{shrunk}}$
Paraná Pesquisas	25	−1.5	**+2.1**
Exame/IDEIA	20	−1.2	+2.3
Veritá	5	+1.9	**+3.8**

The cross-institute spread halves (from approximately 8 pp to 4 pp). The mainstream majors (Datafolha, Quaest, Ipespe) cluster within 1 pp of each other, consistent with their broadly similar methodologies; Veritá's right-leaning posture *strengthens* once industry-wide swings are stripped out; Paraná Pesquisas's small "left bias" under v5 reveals itself as mildly right-leaning under v6+R2.

B.4.6 Estimated election swings **Table HE.3.** Election fixed effects $\hat{\delta}_t$ (pp on share scale). Positive values indicate the industry collectively *underestimated* the challenger; negative values, *overestimated*.

Year	$\hat{\delta}_t$ (pp)	Conjuncture interpretation
2006	−6.95	Industry over-estimated Alckmin
2010	−1.53	Modest over-estimation of Serra
2014	**+5.86**	"Aécio surge" missed in late campaign
2018	**+3.87**	Shy-Bolsonaro voter effect, widely documented
2022	−1.26	Modest over-estimation of Lula

The estimated δ_t aligns with the contemporary reporting of each cycle and provides quantitative confirmation of patterns known qualitatively to election analysts.

B.4.7 Use in OOS forecasting For the target election t^* (whose outcome we wish to forecast), δ_{t^*} is unobserved. We therefore:

1. **Subtract** $\hat{b}_h^{\text{shrunk}}$ from each poll observation (institute-level correction).
2. **Inflate** the irreducible variance by τ_t^2:

$$\sigma^2_{\text{irr}} \;=\; \sigma^2_{\text{drift}}\,\frac{\tau}{7} \;+\; \sigma^2_{\text{house}} \;+\; \sigma^2_{\text{surprise}} \;+\; \boxed{\tau_t^2}\,.$$

This is the central methodological improvement of v6+R2: the unknown δ_{t^*} is *not* set to its expectation (zero); rather, its prior variance contributes to the predictive uncertainty.

B.4.8 In-sample vs out-of-sample trade-off A subtle but important consequence: backtests on the elections used to estimate δ_t will, by construction, show *worse* in-sample Brier under v6+R2 than under v5. The v5 model could "explain" each election's δ_t as institute bias, achieving artificially low residuals; v6+R2 refuses this conflation. The honest assessment is that v6+R2 is the methodologically correct out-of-sample predictor; v5/v6-lean Brier in-sample reflects an upward-biased optimism.

A fully leave-one-out re-estimation of both b_h and δ_t (excluding election t^*) would resolve this asymmetry and is planned for v6.1.

B.5 Validation

B.5.1 Walk-forward backtest For each runoff election

$$t \in \{\,2002,\ 2006,\ 2010,\ 2014,\ 2018,\ 2022\,\},$$

the model is fit using only data available at horizon $\tau = 7$ days before the runoff. Source weights γ_i are applied as specified in §B.2.3, and house effects are estimated from polls in all other elections (leave-one-out style for institute biases).

B.5.2 Brier decomposition (Murphy 1973) The Brier score admits the orthogonal decomposition

$$\text{Brier} \;=\; \text{REL} \;-\; \text{RES} \;+\; \text{UNC},$$

with the components

Term	Meaning	Direction
REL	Reliability (calibration error)	lower is better
RES	Resolution (discriminative power)	higher is better
$\text{UNC} = \bar{y}\,(1-\bar{y})$	Irreducible variance of outcomes	constant per dataset

With 3 wins for the challenger and 3 for the incumbent in our six-election sample, $\bar{y} = 0.5$ and $\text{UNC} = 0.25$.

Table 2. Aggregate metrics across model iterations.

Metric	v4 (no HE)	v5 (HE, 6 sources)	v6.0 (HE, lean)	**v6.1 (recalib.)**	**v6.2 (hier.)**
Brier	0.134	0.121	0.135	**0.132**	0.145
MAE	3.6 pp	3.4 pp	4.5 pp	4.5 pp	4.5 pp
share					

Metric	v4 (no HE)	v5 (HE, 6 sources)	v6.0 (HE, lean)	**v6.1 (recalib.)**	**v6.2 (hier.)**
Correct winners	6/6	6/6	5/6	**6/6**	**6/6**
Mean Pr(actual winner)	0.672	0.656	0.647	**0.683**	0.642

The progression v6.0 → v6.1 → v6.2 reflects deliberate trade-offs. **v6.1** recalibrates $\bar{\gamma}$ to recover the 2010 case (Lula's +55 pp approval peak misforecast under v6.0); 6/6 accuracy with Brier 0.132. **v6.2** adds the hierarchical bivariate decomposition to enable candidate-specific forecasting in 2026; this incurs a calibration penalty (+0.013 Brier) concentrated in 2002 (the only historical case without polls, where the $\sigma_\delta = 1.0$ prior on δ_k inflates posterior variance beyond what approval alone would suggest), while preserving 6/6 accuracy.

Table 2b. Per-election v6.1 vs v6.2 comparison.

Year	Real winner	Pr(A) v6.1	Pr(A) v6.2	$\lvert\Delta\rvert$
2002	A (Lula)	0.679	0.581	0.098
2006	B (Lula inc.)	0.412	0.415	0.003
2010	B (Dilma)	0.500	0.500	0.000
2014	B (Dilma inc.)	0.367	0.370	0.003
2018	A (Bolsonaro)	0.751	0.750	0.001
2022	A (Lula)	0.733	0.733	0.000

The two specifications differ materially only in 2002 — the case with no polls. v6.1 reaches 0.679 on approval alone (FHC's −15 pp net approval with PSDB-Serra as continuity), while v6.2 honors the structural uncertainty of an unanchored δ_k, returning 0.581. Whether this honesty is desirable depends on the use case: for backtest fidelity, v6.1 is preferable; for forward forecasting in 2026 (when several candidates lack polls), v6.2's structural-uncertainty discipline is essential.

B.5.3 Per-election diagnostics **Table 3.** Per-election share and probability estimates (v6.1 recalibrated).

Year	Real winner	s_A^{real}	$\hat{s}_A$ (v6.1)	Pr(A) (v6.1)
2002	A (Lula)	0.613	0.556	0.679
2006	B (Lula inc.)	0.392	0.481	0.412
2010	B (Dilma)	0.439	0.500	0.500
2014	B (Dilma inc.)	0.484	0.471	0.367
2018	A (Bolsonaro)	0.551	0.557	0.751
2022	A (Lula)	0.509	0.552	0.733

The recalibrated weights $(\bar{\gamma}_{\text{polls}}, \bar{\gamma}_{\text{appr}}) = (0.30, 0.70)$ recover 2010 (under v6.0 the model returned 0.50/0.50 because polls in that cycle were neutral despite Lula's +55 pp net approval being a

textbook Time-for-Change signal). 2006 also tightens: with approval correctly weighted, the model assigns 0.412 to challenger Alckmin against the incumbent — close to the actual outcome of 0.392 for the challenger.

The 2010 case remains imperfect: the model returns 0.50 exactly, capturing only that the race was a coin-flip *a priori* but not that Dilma comfortably won 56/44. Lula's +55 pp net approval is the strongest pro-incumbent signal in our dataset; the model captures it as a directional pull but the noisy 2010 polls (38 surveys, $\bar{p}_A \approx 0.50$) anchor the posterior back to neutral. This residual under-confidence is a known limitation; see §B.7.

B.5.4 Sensitivity audit of the approval mechanism The motivation for the v6.1 recalibration was a sensitivity audit, summarized here. For each historical election we compute the local elasticity

$$\frac{\partial \Pr(A \text{ wins})}{\partial A_{\text{avg 3m}}} \text{ (per percentage point)}$$

via centered finite differences (±1 pp), and the ratio of effective precisions $\text{prec}_{\text{polls}}/\text{prec}_{\text{appr}}$ in the posterior.

Table 3b. Operational sensitivity of approval mechanism under v6.0 ($\bar{\gamma}_{\text{appr}} = 0.45$).

Year	Polls n	$\text{prec}_{\text{polls}}/\text{prec}_{\text{appr}}$	$\partial \Pr / \partial A_{\text{avg}}$ (per pp)
2002	0	— (polls absent)	**+1.14%**
2006	9	337	+0.006%
2010	38	**3,112**	+0.001%
2014	15	99	+0.020%
2018	42	2,156	+0.001%
2022	252	**113,669**	+0.000016%

The structural sensitivity ($\alpha_1 = 1.8$ in logit-space, equivalent to a ~ 4 pp share-space response per 20 pp net approval) is **operationally extinguished** in poll-rich elections. Approval was load-bearing only in 2002 (no polls). The 2010 case — the most informative for Time-for-Change — saw the structural signal completely masked by neutral poll precision.

The v6.1 recalibration to $\bar{\gamma}_{\text{appr}} = 0.70$ reduces the polls/approval precision ratio by roughly 3× (e.g., 2010 from 3,112 to $\sim$ 1,040), restoring the approval mechanism to operational relevance.

B.5.5 Drop-one-source sensitivity We zero out each source's γ_i in turn and re-run the backtest under v6.1:

Table 4. Drop-one-source sensitivity for v6.1 (recalibrated lean). Reported Brier values are the model's Brier when source i is forced to weight zero; parenthetical Δ is the change from baseline.

Source dropped	v6.1 Brier (Δ)	Accuracy
(baseline)	0.132	6/6

Source dropped	v6.1 Brier (Δ)	Accuracy
polls	0.072 (−0.060)	6/6
approval	0.157 (+0.025)	**4/6**
market (auxiliary)	0.132 (≡ 0)	6/6
donations (auxiliary)	0.132 (≡ 0)	6/6
roll-call (auxiliary)	0.132 (≡ 0)	6/6
demographics (auxiliary)	0.132 (≡ 0)	6/6

The auxiliary sources mechanically yield $\Delta \equiv 0$ because their $\gamma_i = 0$ (they are out of the posterior by construction). Among the core sources, **approval is now decisively load-bearing**: dropping it loses two elections (accuracy falls from 6/6 to 4/6) and increases Brier by 0.025. This contrasts sharply with v6.0 lean where approval was nominally in the posterior but operationally inert (§B.5.4).

The persistent *negative* Δ for polls (improving Brier by 0.060 when polls are removed) confirms the v6+R2 finding that the institute-level house-effect correction does not fully capture systematic bias — even after recalibrating $\bar{\gamma}$, residual poll bias remains a known source of error. The two-way random-effects extension (§B.4) addresses this in-posterior structurally; the residual reflects the in-sample optimism documented in §B.4.8.

B.5.6 PIT diagnostics The Probability Integral Transform

$$\mathrm{PIT}_t \;=\; F\big(s_{A,t}^{\mathrm{real}} \,\big|\, \mathcal{D}_t\big),$$

where $F(\cdot \mid \mathcal{D}_t)$ is the model's posterior predictive CDF of share s_A given the data, should be approximately Uniform$[0, 1]$ under correct calibration. With $N = 6$ we lack the power for a meaningful goodness-of-fit test, but visually the PIT histogram is consistent with uniformity; the Kolmogorov-Smirnov test yields $p = 0.337$ (cannot reject uniformity).

B.6 Application: 2026 forecast under candidate uncertainty

By May 2026 the PL faces a structural problem: Jair Bolsonaro remains ineligible following the 2023 TSE ruling, and Flávio Bolsonaro — the family's natural heir — became politically encumbered (but not, contra earlier formulations, necessarily withdrawn) by his exposure to the Banco Master crisis (BACEN extrajudicial liquidation Nov 2025; documented family financial entanglements). The succession field has accordingly widened. We model **five** scenarios (v6.4):

1. **Flávio Bolsonaro** (PL) — the *frontrunner*. Author's heuristic: incumbents with cohesive bases tend to resist withdrawal under indictment-grade pressure alone (no systematic literature citation — this is a political reading, not an empirical regularity); we model a −1.5 pp event-shift on his existing polls to capture lag in poll absorption of the Banco Master crisis, *not* a withdrawal probability.
2. **Tarcísio de Freitas** (PL/REP, gov-SP) — *one option* among potential incumbents. Governs the largest electoral college (SP, 22% of population) with a runoff lead in state-level polling; the accept-VP scenario is plausible.

3. **Michelle Bolsonaro** (PL) — symbolic capital and evangelical reach, offset by political inexperience and an author's adjustment for plausible gender drag in candidate substitution (literature on this specific phenomenon in Brazilian elections is thin; Lawless & Pearson 2008 covers a related but distinct dynamic in U.S. primaries).
4. **Caiado** (União/PSD, gov-GO) — dark horse, added in v6.4 because the polling aggregator captured 75 Lula-vs-Caiado head-to-head pairs on Wikipedia-PT (more empirical coverage than Renan, Michelle, or Tarcísio individually).
5. **Renan Santos** (Missão / MBL) — anti-system bloc; weakest structural profile of the five.

We treat this as genuine candidate uncertainty and apply the **law of total probability**:

$$\Pr(\text{right wins}) \;=\; \sum_{k \in \{F,\,T,\,M,\,C,\,R\}} \Pr(\text{right wins} \mid A = k)\;\Pr(A = k),$$

where A denotes the identity of the right-wing candidate and C is Caiado. The succession priors $\Pr(A = k)$ are subjective — derived from public reporting and political-economy literature on family-incumbent retention under judicial pressure — and stated explicitly in Table 5. They are retained in the projection output as counterfactuals.

B.6.1 Candidate-specific inputs (v6.4)

Cand. k	Polls?	Anchor share	$\sigma_{x,k}$	Event shift	Source of anchor
Flávio (F)	yes (57 pairs, $\bar{s} = 0.478 \to 0.463$ post-shift)	—	—	0 (v6.8; −1.5 pp until v6.7)	Banco Master event-shift — deactivated in v6.8, see §B.6.2
Tarcísio (T)	sparse (3 pairs)	0.523	0.20	—	Composite: $0.57 \times 0.22 + 0.51 \times 0.78$ (SP gov 2T + extrapolated non-SP Bolsonaro 2022 transfer)

Cand. k	Polls?	Anchor share	$\sigma_{x,k}$	Event shift	Source of anchor
Michelle (M)	yes (54 pairs)	0.49	0.28	—	Bolsonaro 2022 transfer ($\sim$ 85% evangelical) -3 pp gender drag -1 pp inexperience
Caiado (C)	yes (75 pairs, $\bar{s} = 0.449$)	0.45	0.22	—	Direct aggregation of Wikipedia-PT head-to-head (Nexus/BTG, Meio/Ideia, Real Time, Paraná)
Renan (R)	yes (16 pairs)	0.43	0.35	—	MBL/Novo base (5-10% 1T) + conditional 2T transfer

(Deactivated in v6.8 — see the within-poll test in §B.6.2; this paragraph documents v6.2–v6.7.) The Banco Master shift on Flávio's polls was an *author's adjustment*: scandal-shock absorption in polls is a recognized phenomenon in opinion-dynamics literature, but the specific calibration ("−1.5 pp", "2–4 weeks") is heuristic rather than drawn from a precise published estimate for Brazilian runoff polling. Citations previously claimed for this point (Page & Shapiro 1992 on rational publics; Gelman et al. 2016 on 2016 election lessons) do not in fact substantiate the specific magnitude — they were removed in v6.5 after an audit found them mis-attributed. The state-level polls aggregated by Maynard (2026) were fielded in April 2026, prior to the escalation of the Banco Master investigations into the family's financial ties. We apply a discrete shift of -1.5 pp on the aggregated share for Flávio (from 0.520 to 0.505); the magnitude is deliberately small to remain conservative. Tarcísio's composite anchor is derived from the law of total expectation across SP and non-SP populations:

$$\hat{s}_T \;=\; w_{\mathrm{SP}}\,\hat{s}_T^{\mathrm{SP}} \;+\; (1 - w_{\mathrm{SP}})\,\hat{s}_T^{\neg\mathrm{SP}} \;=\; 0.22 \times 0.57 \;+\; 0.78 \times 0.51 \;=\; 0.523,$$

with $\hat{s}_T^{\text{SP}} = 0.57$ chosen as the midpoint of Datafolha-SP gubernatorial second-round scenarios (Mar 2026: range 50%–60% across opponents — Tebet, França, Haddad, Alckmin; we picked ≈ 0.57 as a deliberate center value), and $\hat{s}_T^{\neg\text{SP}} = 0.51$ obtained by extrapolating from Bolsonaro's 2022 national share (0.491) — note: the "non-SP" sub-share is not directly reported by TSE at the granularity needed; we use the national share as approximation plus a +2 pp adjustment for center-right voters who reject the family but accept Tarcísio. The $\sigma_x = 0.20$ (in logit-space, $\approx$ 5 pp share) reflects this approximation uncertainty. (See above for full anchor derivation; $\sigma_x = 0.20$ reflects the approximation uncertainty.)

B.6.2 Per-scenario posteriors (v6.4 hierarchical) Priors revision (v6.4, 2026-05-27; honest framing in v6.5). The earlier hierarchy (Tarcísio 0.45, Flávio 0.15) treated Tarcísio's emergence as the presumed nominee with what we now consider over-confidence. We re-balance the priors to recognize Flávio as the *frontrunner* — albeit with a −1.5 pp event-shift still applied to capture lag in poll absorption of the Banco Master crisis — and treat Tarcísio as *one option among potential incumbents*. We label this an **author's prior, not a citation-backed empirical regularity**; an internal audit (2026-05-27) found that citations earlier offered for this re-weighting (Page & Shapiro 1992; Gelman et al. 2016 — see References) were mis-attributed and have been removed. We also add **Caiado** as a new scenario: the polling aggregator captured 75 head-to-head pairs of Lula vs. Caiado on Wikipedia-PT (more empirical coverage than Renan, Michelle, or Tarcísio individually), making it methodologically indefensible to exclude.

Table 5. Succession priors and per-scenario conditional probabilities (v6.9). Two intervals are reported, because they answer different questions. *95% interval of* Pr: the uncertainty in the win probability itself that more data could reduce — $\varphi_s \sim \mathcal{N}(\mu, \sigma_{\text{lik}}^2)$ mapped through $\Phi(\varphi_s/\sigma_{\text{irr}})$, whose mean is exactly $\Pr(A \text{ wins} \mid k) = \Phi(\mu/\sigma_{\text{total}})$. *95% predictive interval of share*: A's two-party vote share, $\text{expit}(\mu \pm 1.96\,\sigma_{\text{total}})$.

Successor k	Prior $\Pr(A = k)$	$\Pr(A \text{ wins} \mid k)$	95% interval of Pr	Median share	95% predictive interval of share
Flávio Bolsonaro (F)	1.00	**0.480**	[0.38, 0.58]	0.495	[0.30, 0.69]
Tarcísio de Freitas (T) — counterfactual	0	0.577	[0.25, 0.87]	0.523	[0.30, 0.73]
Michelle Bolsonaro (M) — counterfactual	0	0.441	[0.35, 0.54]	0.484	[0.29, 0.69]

Successor k	Prior $\Pr(A = k)$	$\Pr(A \text{ wins} \mid k)$	95% interval of Pr	Median share	95% predictive interval of share
Caiado (C) — counter-factual	0	0.374	[0.29, 0.47]	0.465	[0.27, 0.67]
Renan Santos (R) — counter-factual	0	0.236	[0.17, 0.32]	0.423	[0.24, 0.63]

Interval correction (v6.7). Up to v6.7 the column labelled "IC95%" in this table — and every `P_A_wins_ci95_*` field in the projection output — was the predictive interval of the vote *share* (percentiles of expit(φ)), presented as if it were an interval of the win *probability*. The two are different quantities, which is why a point estimate could fall outside its own "interval" (Renan: Pr = 0.235 against [0.24, 0.63]). The point estimates were always correct; only the interval was mislabelled. Flávio's previously published "[0.29, 0.69]" is his share interval; the interval of his win probability was [0.37, 0.57] at the time ([0.38, 0.58] after the v6.8 change).

Five candidates now receive **distinct** posterior probabilities. Caiado's lower $\Pr(A \text{ wins} \mid C) = 0.374$ reflects the empirical head-to-head share across 167 pairs (mean $\bar{s}_C = 0.462$), translated through the bivariate posterior. Flávio's 0.480 reflects 170 head-to-head pairs whose mean share sits just below parity (0.492), with no Banco Master shift since v6.8.

Marginal result. Combining the conditionals with the succession priors yields

$$\Pr(\text{right wins 2026 runoff}) = 0.480, \qquad \Pr(\text{Lula}) = 0.520.$$

The decomposition of the marginal:

Successor k	$\Pr(\text{right} \mid k) \times \Pr(A = k)$
Flávio	$\mathbf{0.480 \times 1.00 = 0.480}$
Tarcísio	$0.577 \times 0 = 0$
Michelle	$0.445 \times 0 = 0$
Caiado	$0.365 \times 0 = 0$
Renan	$0.235 \times 0 = 0$
Total	0.480

Comparison to v6.2 and v6.6. The v6.2 marginal was 0.502 under priors $(T, M, F, R) = (0.45, 0.25, 0.15, 0.15)$; v6.6 published 0.424 under the v6.4 priors $(F, T, M, C, R) = (0.40, 0.25, 0.15, 0.10, 0.10)$. The v6.7 marginal equals the Flávio conditional because the succession is resolved: the official registrations leave Flávio as the right's only viable runoff candidate — Tarcísio (still governor of São Paulo) and Michelle are not running, and Caiado

and Renan poll at or below 3% in the first round (Quaest, 2–3 October: Lula 46%, Flávio 45% of valid votes). The other four scenarios remain computed as counterfactuals in the projection output.

Effect of the v6.6 vantage-point correction. Re-running the marginal across $\tau \in \{150, 120, 90, 60, 54, 30, 7\}$ gives $0.441, 0.437, 0.431, 0.424, 0.424, 0.423, 0.418$ — monotonically *down*, which is the opposite of what the recency intuition suggests. Two mechanisms compound. First, the newly admitted window is not uniformly favourable to the right: Flávio's monthly mean $\bar{s}_F$ runs 0.506 (March), 0.475 (June), 0.476 (July), 0.486 (August), 0.499 (September) — his trough falls squarely inside the previously excluded window, and the pool mean over all 130 pairs (0.4867) is *below* the mean over the 41 pairs that survived $\tau = 150$ (0.4919). Second, a shorter τ shrinks $\sigma^2_{\text{drift}} \propto \tau$, so the posterior is pulled less toward 0.5; with $\bar{s}_F < 0.5$ this moves $\Pr(\cdot \mid F)$ further below parity. The same mechanism runs the other way for Tarcísio, whose anchor sits above parity and who therefore *gains* (0.558 → 0.571). The correction is nonetheless the right one: a forecast published in September should not be standing at a June vantage point.

Correction in v6.7. The sweep above was computed with a 28-day misalignment: runoff-pair `date_dte` was counted to the first round (4 October) while `tau_days` counts to the runoff (1 November). On 8 September the filter therefore cut at 11 August, not "today" as v6.6 stated; on 4 October it was still discarding the 62 most recent Flávio pairs. Two defects are fixed in v6.7, with their effect on $\Pr(\cdot \mid F)$ at 4 October: (i) runoff-pair `date_dte` is now counted to the runoff, as in the historical data: 0.396 → 0.425 (181 instead of 119 pairs admitted); (ii) the −1.5 pp Banco Master shift — whose stated purpose was to correct the *lag* of polls fielded before the escalation of the case — was being applied to every poll, including June–October polls that already reflect the scandal; it is now limited to pairs fielded before 15 May 2026: 0.425 → 0.471. The result is robust to the cutoff (0.472 for 1 May, 0.467 for 1 June). The September approval update (−8.0 → −8.5 pp; Datafolha, AtlasIntel and Quaest average) has no material effect.

Banco Master shift removed (v6.8). The −1.5 pp shift assumed that polls fielded before the scandal overstate Flávio's *current* support. We tested that directly by comparing Flávio with Caiado *within the same poll* (same institute and date), which removes house effects and the shared Lula baseline. Flávio's lead over Caiado fell from +3.9 pp (polls before 1 May, $n = 29$) to +2.0 pp (mid-May to July, $n = 38$), while Caiado's own share was flat — a Flávio-specific dip, consistent with the scandal. But it was temporary: in September–October Flávio is back at 0.50–0.51 of valid votes, his pre-scandal level, and his lead over Caiado (+2.3 pp, $n = 62$) matches December–January (+2.2/+2.5 pp); February–April was his peak, not his baseline. Pre-scandal polls therefore do *not* overstate his current support, and the shift was pulling him below where the data place him. Removing it moves $\Pr(\cdot \mid F)$ from 0.471 to 0.481. The mechanism (`event_shift_pp`, `event_shift_until`) remains in the code, deactivated; reactivating it should require evidence of a persistent effect.

Data quality control (v6.9). A pass over the published dashboard found three ingestion errors, all fixed. (i) Wikipedia-PT often gives fieldwork periods without a year ("13 Ago – 17 Ago"); the parser assumed the election year and only stepped back a year if the date fell in the future. Once August 2026 had passed, 2025 polls (January–October) were stamped as 2026 — e.g. Quaest's August 2025 runoff poll (Lula 48, Flávio 32) appeared as 17 August 2026. Years

now come from the table's section heading, with a month-sequence wrap rule for the first-round tables that are sectioned by month only. (ii) Tables under "Agregação de pesquisas"/"Polling aggregation" (UOL, PollingData/BBC, The Economist, Agregador presidencial, Plano Político, Poll+Trend, CartaCapital, ABC Dados) are averages of other polls; 11 such rows, all in the last two weeks, had entered the pool as if they were polls, double-counting respondents where the drift weight is highest. They are now skipped. (iii) The Quaest 2–3 October runoff row on Wikipedia-PT has the two candidates swapped (42/44); the poll's own report gives Lula 44, Flávio 42. A documented correction layer (`data/polls/manual_corrections.csv`) now applies such fixes with source and reason. The Flávio pool goes from 181 to 170 pairs and $\Pr(\cdot \mid F)$ from 0.481 to 0.480 — the errors largely offset. The within-poll Banco Master test was re-run on the cleaned data: the May–July dip remains (it was real, not an artefact of the misdated 2025 polls) and so does the recovery.

Matchup-specific Lula shares (v6.9 data fix). The live poll file was deduplicated on (pollster, date, candidate), which collapsed two things that must stay separate: the first- and second-round values of the same poll, and Lula's share across the different runoff tables of the same poll (Lula×Flávio, Lula×Caiado, …), where it differs. The key now includes the round (from the section heading) and the matchup. The Flávio pool is unaffected — his table comes first in every section, so the retained Lula value was already the right one — but most counterfactual pairs had Lula's share from another matchup (Caiado 98 of 167, Michelle 52 of 61, Renan 70 of 96). Corrected conditionals: Caiado 0.374, Michelle 0.441, Renan 0.236. The Lula × Tarcísio tables are still not recognised (the header "Tarcísio Republicanos" is not in the name map), so the Tarcísio counterfactual remains anchor-only.

Transfer sensitivity. Rejection of Flávio *by voters of other candidates* is not published. The closest proxy is Quaest's 24–27 September runoff crosstab by first-round vote (Flávio/Lula/uncommitted): Zema 81/9/10, Caiado 42/30/28, Renan 48/23/29, Cury 34/27/39. Committed members of these groups are already counted in the head-to-head polls; only the uncommitted share is new information. Allocating it in proportion to each group's committed split — with each group's size taken from the first-round poll average (v6.10; until v6.9 an *assumed* 2.25% each) — moves Flávio's share by +1.01 pp and $\Pr(\cdot \mid F)$ to 0.518. We report this as a sensitivity rather than in the headline: the adjustment cannot be backtested and it partially double-counts the head-to-heads.

State tickets, coalitions and the first-round bridge (v6.10). We tested whether gubernatorial races — where left and right are thought to consolidate — tell us where the voters of eliminated presidential candidates go. Using TSE returns aggregated by state (2002–2022) and the registered coalitions, each governor candidate is classed as aligned with the PT nominee, with the runoff opponent, or neither (`scripts/analysis_governor_blocs.py`). With the presidential first round known, adding the state left–right gubernatorial bloc to a state-level regression of the PT runoff share reduces RMSE by at most 0.1 pp in any election, with coefficients of either sign around zero; on its own the bloc misses by 7–17 pp. Ticket-splitting is massive: in 2022, left-aligned governor candidates took 24% nationally and right-aligned ones 50%, against Lula's 48% in the presidential first round. Leave-one-election-out, a transfer rate predicted from the gubernatorial surplus misses the national runoff by 4.36 pp RMSE versus 4.01 pp for the historical mean transfer rate. Coalition size (seats won by the coalition's parties at the previous election, the basis of free TV time) is collinear with incumbency — the larger

finalist coalition was always the government's and won 4 of 5 — and adds nothing to approval. None of these enters the model. Two first-round outputs are now reported outside the headline: the first-round average (latest poll per pollster within 10 days, 16 pollsters, valid votes: Lula 44.5, Flávio 42.9, Cury 3.7, Renan 3.6, Caiado 3.3, Zema 1.0) and a bridge that keeps Lula's and Flávio's voters and splits the four minor electorates by the average of the first-to-second-round conversion grids of Genial/Quaest (mais recente) and Paraná Pesquisas (range across grids, also including Real Time Big Data July and Quaest 24–27 September: 50.3–51.8%), giving Flávio 51.2% of the runoff valid vote versus 49.5% from the head-to-heads. The transfer sensitivity now uses the measured group sizes instead of the assumed 2.25%, and the same averaged grids, whose uncommitted shares are larger than Quaest's September ones: $\Pr(\cdot \mid F) = 0.518$. This assumes voters who answer blank/null/undecided in the runoff question end up voting like their group's committed voters, which is generous.

B.6.3 Stress on Lula's approval Shocking the average net-approval input by $\Delta A \in \{-0.20, -0.10, 0, +0.10, +0.20\}$ under v6.4 (5 scenarios):

ΔA	$\Pr(F)$	$\Pr(T)$	$\Pr(M)$	$\Pr(C)$	$\Pr(R)$	Pr(right, marginal)
−20 pp	0.481	0.587	0.442	0.375	0.237	0.481
−10 pp	0.481	0.582	0.441	0.374	0.237	0.481
0 (baseline)	0.480	0.577	0.441	0.374	0.236	0.480
+10 pp	0.480	0.572	0.440	0.373	0.236	0.480
+20 pp	0.479	0.567	0.440	0.373	0.236	0.479

Total sensitivity across ±20 pp of approval shock is approximately ±0.1 pp on the marginal (even smaller than v6.2's ±1.4 pp), reflecting the empirical mass that polls now carry — Flávio has 181 head-to-head pairs at 28 days from the runoff, so polling is informationally saturated. The reduction is structural, not a bug: under the bivariate hierarchical model, polls and anchors observe the **total** $\varphi_k = \varphi_{\text{base}} + \delta_k$, so a counterfactual shift in macro mood (approval) is partially absorbed by an offsetting movement in δ_k through the posterior covariance term (§B.2.5.3). The interpretation: *given fixed polls and anchors, shocking approval is informationally coherent only to the extent that the data have not already revealed the macro mood.* In a world where polls are abundant and recent (as in the historical record), shifting approval ex post is a thought experiment that the model cannot reconcile with the observed polls.

For a *pure* approval stress (interpreted as "what if approval changed and polls/anchors had not yet captured it?"), the appropriate experiment is to additionally drop polls and anchors from the posterior, leaving only approval. That stress yields a ±20 pp range on Pr(right) of $[0.42, 0.58]$ — closer to v6.1's response but with the explicit caveat that it ignores all candidate-specific information.

B.6.4 Stress on succession priors Holding the conditionals fixed and assigning all prior mass to a single successor:

Successor with $\Pr(A = k) = 1$	Pr(right)
Tarcísio	0.577
Flávio	0.480
Michelle	0.441
Caiado	0.374
Renan	0.236

Until the registrations resolved the succession, the identity of the right's candidate mattered materially, and the counterfactual pure-candidate worlds still span 34.1 pp (Tarcísio 0.577 vs. Renan 0.236). Caiado sits in the lower-mid range at 0.374 — head-to-head polls show him competitive in the 40 pp band but not crossing the 50 pp threshold against Lula. Since v6.7 the marginal *is* the Flávio world (0.480 in v6.9); the table is retained as a counterfactual.

B.7 Limitations

Honest enumeration of what remains imperfect:

0. **The "6/6" claim is structurally weaker than headline suggests (audit 2026-05-27).** An internal data-quality audit found that the historical backtest leans on two unequal regimes:
   - **2014 / 2018 / 2022** have proper second-round opinion polls in `data/polls/polls_historicas.csv` (104, 50, 1862 rows respectively); the model fuses polls with approval here.
   - **2002 / 2006** have *zero* second-round polls in the historical file; the model relies entirely on approval + fundamentals for those two elections. The "polls 2T" file for 2006/2010 contains per-state election results (UFs in the `date_field` column), which leak the target into the fit if naïvely treated as polls. The fit-evaluation is also **in-sample**: $\bar{\gamma} = (0.30, 0.70)$, $\alpha_1 = 1.8$, and the cap $S_2 = 30$ were calibrated against the same six elections used for evaluation. The honest backtest is $N = 3$ (Brier on 2014/2018/2022 ranges 0.07−0.20 depending on election), with three more cases serving as plausibility checks under fundamentals-only fits.
1. **Sample size**. With $N = 6$ historical runoffs (only $N = 3$ strongly informative — see note 0), all aggregate metrics carry bootstrap IC95% widths of approximately ±0.06 on Brier. The model is best understood as a structured prior informed by historical patterns rather than a high-confidence predictor.
2. **Expert anchors are the brittleness layer of v6.2.** Adding the hierarchical decomposition shifted the locus of subjectivity from *succession priors* (the $\Pr(A = k)$ weights) to *anchor inputs* (the share-space estimates and their uncertainties σ_x). We mitigate by anchoring with explicit empirical compositions wherever possible (Tarcísio = SP gov + extrapolation), but the Michelle and Renan anchors remain primarily structural (literature-calibrated rather than data-driven). A worthwhile extension is to estimate σ_x from historical analogous substitutions (e.g., 1989 Maluf → Colônia, 2018 Lula → Haddad).
3. **House-effect inhomogeneity** (partially addressed). The v6+R2 two-way decomposition captures election-wide swings δ_t but still assumes b_h is *constant in time*. A leave-one-out re-estimation of b_h within v6.2 is planned for v6.3.
4. **MRP not implemented**. The demographic source enters as a placeholder. Full imple-

mentation requires PNAD-C microdata beyond the open SIDRA endpoint. The SIDRA tables for joint distributions over education × income × race × urbanization exist (7448, 7449) but were not pulled for v6.2; expected improvement is ≤ 0.01 Brier with ~ 1 day of engineering.

5. **Pre-2002 elections excluded**. 1998 has no runoff (FHC won outright in the first round); poll coverage for 2002 runoff was not extracted by our parser. 2006 has only 9 polls.
6. **Market data shallow**. Prediction-market liquidity is substantive only for 2022. For 2026, Polymarket has thin order books for most non-Flávio scenarios — limiting the diagnostic value of market signals in differentiating the five-candidate succession field (Flávio, Tarcísio, Michelle, Caiado, Renan).
7. **Approval series compiled**. Quarterly approval series is compiled from publicly reported peaks/troughs of Datafolha/CNT/ Ipsos/Quaest rather than a single source's longitudinal data. Substituting a single-source quarterly series would tighten the analysis.
8. **2018 attribution**. Treating Haddad (PT) as the "incumbent/governist" is symbolic: the legal incumbent was Temer (MDB), who was not on the ballot. The substantive impact on the 2018 prediction is small but the attribution should be revisited for finer-grained simulations.
9. **Approval stress under v6.2 has a non-obvious interpretation.** The hierarchical bivariate model coherently couples polls and approval through the posterior covariance; a ceteris paribus approval shock therefore moves the marginal probability less than it would under v6.1 flat. We document a complementary "pure approval" stress (drop polls and anchors) in §B.6.3 for readers who want the v6.1-style elasticity.

B.8 Conclusion

The model presented yields well-calibrated, interpretable forecasts for Brazilian presidential runoffs from a parsimonious two-source core under a closed-form Gaussian posterior in logit space. Successive revisions have produced a coherent specification:

- **v6.0**: lean two-source core with house effects.
- **v6.1**: recalibrated $\bar{\gamma}$ to recover the 2010 Time-for-Change case (6/6 accuracy, Brier 0.132).
- **v6.2**: bivariate hierarchical decomposition $\varphi_k = \varphi_{\text{base}} + \delta_k$ enabling candidate-specific forecasts even without dedicated polls.

The decision to focus on the runoff — rather than the multi-candidate first round — simplifies the mathematics enough that no MCMC is required, while preserving the substantively important question of who becomes president. House-effect correction via two-way partial pooling materially improves accuracy on the hardest cases (2010, 2022) and isolates election-wide swings δ_t from persistent institute bias b_h. The hierarchical extension restores the ability to distinguish candidates without runoff-specific polls — at a deliberate calibration cost of +0.013 Brier in-sample, accepted because the alternative ("candidate-blind" v6.1) misrepresents the structural variation that genuinely exists across the 2026 succession field.

Applied to 2026, the v6.10 framework (superseded by v7.0, Sections 1–9) produces Pr(right wins 2026 runoff) = 0.480, Pr(Lula) = 0.520 (data refresh 2026-10-04, first-round day; Wikipedia polls through 4 October) — a statistical tie with Lula marginally ahead, conditional on a runoff taking place — under the current premise stack:

- Aprovação líquida Lula: −8.5 pp (September 2026; Datafolha, AtlasIntel and Quaest average);
- Succession: resolved by the official registrations — prior on Flávio = 1 (the other scenarios remain computed as counterfactuals);
- Banco Master event-shift: none (removed in v6.8 after a within-poll test; −1.5 pp in v6.2–v6.7);
- Tarcísio composite anchor: 0.523 ($\sigma_x = 0.20$);
- Forecast vantage point (v6.10): $\tau = 28$ days to a runoff wrongly dated 1 November (actual date: 25 October; corrected in v7.0) — all 170 Flávio pairs published to date (aggregator averages excluded) are admitted, and σ_{drift} is scaled to the remaining campaign time (v6.6, aligned in v6.7);
- Caiado anchor: 0.45 ($\sigma_x = 0.22$), informed by 167 head-to-head pairs on Wikipedia-PT;
- Michelle anchor: 0.49 ($\sigma_x = 0.28$);
- Renan anchor: 0.43 ($\sigma_x = 0.35$).

With the succession resolved, the remaining uncertainty is the Flávio–Lula runoff itself, which the polls place within the margin of error; the forecast is weakly sensitive to ceteris paribus approval shocks (±0.1 pp on the marginal), reflecting the hierarchical model's coherent coupling of polls and approval. For a pure-approval counterfactual, see §B.6.3.

The v6.4 specification represents the model's current best balance between in-sample fidelity (where v6.1 flat slightly edges it), structural identifiability (where the hierarchical decomposition is essential), and forecasting honesty across the five-candidate 2026 field (now including Caiado). It is best understood as a structured prior over election outcomes informed by the historical six-runoff record, transparent about its sources of uncertainty, and explicitly auditable in each of its components. The succession priors specifically are exposed as interactive controls in the dashboard — a reader who disagrees with our weighting (e.g., one who believes Flávio's withdrawal is more likely) can re-run the marginal in seconds.

B.9 References (v6)

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Working papers and grey literature

- Maynard, L. (2026). *Brasil Macro — Cenários Presidenciais 2026*. Paragon Capital working paper. https://rpubs.com/lucasmaynard/1418609

B.10 Data sources (v6)

The model is built on publicly available data. Where the original source imposed access friction (e.g., the TSE bulk-download portal timed out during compilation in May 2026), we mirrored the figures through secondary aggregators with explicit cross-checks against primary sources.

Electoral results and candidacies

- **Tribunal Superior Eleitoral (TSE) — Open Data Portal.** https://dadosabertos.tse.jus.br/. Primary source of certified Brazilian presidential election results (1998–2022) and candidate registries. Mirrored via Wikipedia election tables for this paper with manual audit of 2018 results.
- **Wikipedia — *Eleição presidencial no Brasil em* ⟨*YYYY*⟩.** Used as a curated mirror of TSE results and as a source of consolidated opinion-poll tables. Each yearly article aggregates primary poll filings (TSE BR-NNN/YYYY registrations).

Opinion polls The runoff-poll dataset (356 polls, 2006–2022) was scraped and normalized from the Wikipedia article *Pesquisas de opinião para a eleição presidencial no Brasil em* ⟨*YYYY*⟩ and the *Eleição presidencial* article (2010 polls were embedded in the parent article). Each poll carries a TSE registration code (BR-NNNNN/YYYY), enabling cross-validation against the primary filings if needed.

Institutes covered: Datafolha, IBOPE / Ipec, Vox, Ipespe, Atlas, Quaest (Genial/Quaest), Real Time Big Data, Paraná Pesquisas, CNT/MDA, Futura Inteligência, Instituto Veritá, Modal, FSB, Ipsos, Sensus, Exame/IDEIA, PoderData.

Government approval A quarterly approval series (1998 Q3 to 2026 Q2, 78 data points) was compiled from published peaks/troughs of the following organizations: **Datafolha**, **CNT/MDA**, **Ipsos**, **Genial/Quaest**. The underlying source pages, archived via the Wikipedia *Governo* ⟨*Presidente*⟩ articles, are:

- https://pt.wikipedia.org/wiki/Governo_Fernando_Henrique_Cardoso
- https://pt.wikipedia.org/wiki/Governo_Lula
- https://pt.wikipedia.org/wiki/Governo_Dilma_Rousseff
- https://pt.wikipedia.org/wiki/Governo_Michel_Temer
- https://pt.wikipedia.org/wiki/Governo_Jair_Bolsonaro
- https://pt.wikipedia.org/wiki/Governo_Lula_III

Electoral donations

- **TSE — Prestação de Contas**. https://dadosabertos.tse.jus.br/dataset/?groups=prestacao-de-contas. Total nominal donations (party fund + electoral fund + private donations) by candidate, 2002–2022. Top-four presidential candidates per cycle were compiled into `data/tse/donations_real.csv`.

Legislative roll-call votes

- **Senado Federal — Dados Abertos Legislativos.** https://legis.senado.leg.br/dadosabertos/plenario/lista/votacao/$/langle$YYYYMMDD$/rangle$/$/langle$YYYYMMDD$/rangle$. 1,767 plenary votes, 142,267 individual senator votes (2011–2025). Used to estimate coalition cohesion c_p (Section 2.2.5).
- **Câmara dos Deputados — Dados Abertos.** https://dadosabertos.camara.leg.br/api/v2/votacoes. 359 plenary votes from the 56th and 57th legislatures (2019–2026).
- **Câmara dos Deputados — Relatórios da Atividade Legislativa.** https://www2.camara.leg.br/atividade-legislativa/plenario/relatorios-da-atividade-legislativa/dados-de-votacoes-nominais. Reference URL provided by the project's specification document.

Demographics, economics

- **IBGE — Sistema SIDRA.** https://apisidra.ibge.gov.br/. PNAD-C and Census 2022 tables (escolaridade 6407, rendimento 7434, desocupação 6318, população 4709, PIB per capita 5938, urbanização 1378).
- **IBGE — *Pesquisa Nacional por Amostra de Domicílios* (PNAD-C).** Reference page: https://www.ibge.gov.br/estatisticas/sociais/populacao/9127-pesquisa-nacional-por-amostra-de-domicilios.html.
- **IBGE — Censo Demográfico 2010, 2022.** Tables 137 and 200 (population by federation unit), table 1378 (urbanization rates).

Prediction markets

- **Polymarket — Gamma API.** https://gamma-api.polymarket.com/markets. Historical pricing for the contract "Lula wins 2022 Brazil election" and emerging 2026 contracts.
- **Kalshi — Trade API.** https://api.elections.kalshi.com/trade-api/v2. No liquid 2022 contracts; partial 2026 coverage.

State-level 2026 polls (aggregator)

- Maynard, L. (Paragon Capital). *Brasil Macro — Cenários Presidenciais 2026*. https://rpubs.com/lucasmaynard/1418609. Aggregator of state- level second-round polls for the Flávio Bolsonaro × Lula scenario (12 federation units, ≈ 79% of population, April–May 2026). Includes the qualitative observation of Veritá's right-leaning bias in SC 2022 (Lula 25.6 % poll vs 30.7 % real, −5.1 pp residual) used as a sanity check on §B.4 estimates.

Project-internal artifacts All processing scripts and intermediate CSVs are version-controlled in the project's research repository (see Appendix A for the directory listing). Each ingestion script documents its endpoint and the date of retrieval (May 20–21, 2026).

Appendix C — Updates after the pre-registration (8–16 October 2026)

Sections 2.4, 2.6 and 6 describe the model as fixed on 6 October 2026, before any post-first-round poll. This appendix records what happened when the polls arrived. Nothing in the model was changed: the calibration (`results/v7_calibration.json`), $\sigma_F(h)$, the house term ($\sigma_1 = 0.0696$, $k_{\text{ref}} = 2.83$), the bias $b = 1.31$ pp and the divergence rule are the ones in §2. Only data came in, and three processing errors were fixed (§C.5); none of them touches the headline number.

C.1 The post-first-round polls

Seventeen runoff polls from 13 pollsters were published between 8 and 16 October. Wikipedia had not yet listed any of them, so they were entered by hand from the pollsters' reports or the press, with the TSE registration and the source URL (`data/polls/manual_polls_2026.csv`). Total-vote figures are used, as in the historical series; the model takes the two-party share.

Pollster	Fieldwork	n	Flávio	Lula	Flávio, two-party
PoderData/Aya	5–7 Oct	3,000	49.0	44.0	52.69
Vox Brasil	5–7 Oct	2,100	42.7	44.2	49.14
Datafolha	6–7 Oct	2,520	49.0	45.0	52.13
AtlasIntel/Bloomberg	3–8 Oct	5,026	51.1	45.7	52.79
Ipespe/Abrapel	6–8 Oct	1,500	49.0	44.0	52.69
DataTrends	7–9 Oct	2,000	50.0	44.0	53.19
Futura	7–10 Oct	2,000	50.0	46.6	51.76
Gerp	8, 9, 12 Oct	2,400	53.0	44.0	54.64
Nexus/BTG	10–12 Oct	2,002	48.0	45.0	51.61
PoderData/Aya	10–12 Oct	3,000	48.0	46.0	51.06
Palver	10–13 Oct	5,000	50.0	49.0	50.51
Quaest	10–13 Oct	2,154	46.0	44.0	51.11

Pollster	Fieldwork	n	Flávio	Lula	Flávio, two-party
Datafolha	13–14 Oct	2,520	47.0	45.0	51.09
Indexa/Broadcast	13–14 Oct	2,000	44.0	42.0	51.16
CNT/MDA	13–15 Oct	2,002	46.8	45.9	50.49
Futura	14–15 Oct	2,000	50.8	49.2	50.80
PoderData/Aya	13–15 Oct	3,000	47.0	47.0	50.00

Notes on the inputs:

- AtlasIntel's fieldwork began on 3 October, the day before the first round. Under the fixed rule (end date after the first round) it is a post-first-round poll; the share of interviews taken before the vote is not published.
- Futura 14–15 October is published only in valid votes; the PoderData/Aya totals for 13–15 October come from one outlet (SBT News); CNT/MDA's sample is reported as 2,014 by one source and 2,002 by another (2,002 is used).
- Palver is an opt-in online panel with a design effect of 2.8 (effective $n = 1{,}783$). The pooled average weights by nominal n, as for every other poll; the effective-n variant is in §C.4.

C.2 The three updates

All quantities as defined in §2. $\bar{p}$ is the n-weighted mean of the post-first-round two-party shares (in logit, back-transformed); h is days to the runoff; z is the divergence statistic of §2.6.

	6 Oct (pre-reg.)	8 Oct	9 Oct	16 Oct
Post-first-round polls / pollsters k	0 / 0	1 / 1	2 / 2	17 / 13
Raw average $\bar{p}$	—	52.13%	52.57%	51.58%
Corrected $(\bar{p} + b)$	—	53.44%	53.88%	52.89%
h (days)	—	17	16	9
$\sigma_F(h)$ / $\sigma_{F,\text{eff}}$	—	0.138 / 0.149	0.132 / 0.135	0.100 / 0.100
Weight: polls after / bridge / polls before	— / 0.93 / 0.07	0.65 / 0.33 / 0.03	0.79 / 0.19 / 0.02	0.92 / 0.07 / 0.01
z (alert if $\lvert z\rvert > 2$)	—	+0.34	+0.50	+0.17
$\hat{s}$ (Flávio share)	52.2%	53.0%	53.5%	52.8%
$\hat{\sigma}$ (logit)	0.166	0.141	0.132	0.099
95% interval	44.1–60.3%	46.1–59.8%	47.1–59.9%	48.0–57.6%
Pr(Flávio)	**0.705**	**0.805**	**0.859**	**0.875**
Pr(Flávio), t_4 tails	—	0.854	0.899	0.910

With $k = 13 > k_{\text{ref}}$ the house term of §2.4 is zero and the polls carry the tabulated $\sigma_F(h)$. The

bridge (52.27%) did not move; the pre-first-round component moved from 51.93% to 52.14% because Wikipedia added late pre-first-round polls (Gerp, Veritá).

Check against the pre-registration. The table of §6.3, computed on 6 October for two pollsters 16 days out, gives Pr(Lula) = 0.17 at a raw average of 52% and 0.12 at 53%; interpolated at the 9 October average (52.57%), 0.14. The model gave 0.141. The mechanics behaved as announced.

Divergence. The alert never fired. On 9 October the pre-registered band for two pollsters was 44.5–57.4% of the raw two-party share; the average was 52.6%. On 16 October $z = +0.17$: polls and bridge agree to within a fifth of a standard deviation.

C.3 Why the probability rose while the share barely moved

From 6 to 16 October the central estimate moved by 0.6 points (52.2% → 52.8%), but $\hat{\sigma}$ fell from 0.166 to 0.099 in logit (about 4.1 to 2.5 points). The standardised lead $\text{logit}(\hat{s})/\hat{\sigma}$ went from 0.54 to 1.15. The gain is in precision, not in the level: post-first-round polls nine days out have $\sigma_F = 0.100$ against $\sigma_B = 0.168$ for the bridge, and the common drift left to run shrinks from 21 to 9 days.

C.4 Sensitivity on 16 October

The fixed rule pools every post-first-round poll with weight $\propto n$ and no recency weighting. The series moves towards Lula: polls ending 5–10 October average 52.18% raw (7 polls), those ending 11–15 October 51.16% (10 polls). Variants (none replaces the headline):

Variant	Raw average	$\hat{s}$	Pr(Flávio)	z
Pre-registered rule (all 17)	51.58%	52.84%	0.875	+0.17
Latest round per pollster (13)	51.47%	52.74%	0.866	+0.14
Fieldwork ending on or after 12 Oct (10)	51.16%	52.45%	0.839	+0.05
Without AtlasIntel	51.42%	52.70%	0.862	+0.13
Without Gerp and Vox Brasil (extremes)	51.52%	52.79%	0.870	+0.15
Palver at effective n = 1,783	51.66%	52.92%	0.881	+0.19

Other checks: splitting the grids' undecided evenly in the bridge gives Pr(Flávio) = 0.873; varying the own-error floor (5–50%) and σ_d across its 90% interval moves Pr(Lula) between 0.116 and 0.125.

Holding everything else fixed, with 13 pollsters, Pr(Lula) as a function of the raw post-first-round average:

Raw average, Flávio two-party	Pr(Lula), 16 Oct ($h = 9$)	Pr(Lula), 22 Oct ($h = 3$)
48%	0.57	0.53
49%	0.43	0.38
50%	0.29	0.24
51%	0.18	0.14
52%	0.10	0.07
53%	0.05	0.03

Pr(Lula) = 0.5 requires a raw average of 48.5% for Flávio on 16 October (48.2% three days out): a shift of about three points from the current average.

Relation to the tail study (§7). The study's defensible range, 0.26–0.32, was computed on 6 October from the first-round result alone. The published Pr(Lula) = 0.125 now lies below it. This is not a contradiction: the study had no post-first-round polls, and seventeen of them have since narrowed the uncertainty around a Flávio lead.

C.5 Processing corrections

1. **First-round poll window.** `first_round_average` averaged polls in a 10-day window ending *today*. After 14 October the window held no first-round polls and the projection script failed. The window now ends on first-round day. It feeds only diagnostics (first-round poll error, critical defection, the v6 transfer sensitivity); the v7 bridge uses the TSE count. The window is now 24 September to 4 October; in the 9 October run it was 29 September to 9 October, and the first-round poll error reported there moves from +2.1 to +2.0 pp.
2. **Manual × Wikipedia de-duplication.** A hand-entered poll was dropped only if Wikipedia listed the same pollster name and end date. Wikipedia spells pollsters several ways (PoderData vs PoderData/Aya, Nexus/BTG vs BTG/Nexus) and sometimes shifts the end date by a day, so the same poll could have entered twice. A manual row is now dropped if the candidate and round match, the pollster names share a token and the end dates are at most two days apart (with a test). Since 17 October the manual row, checked against the pollster's report, also *replaces* the Wikipedia row for the same poll: Wikipedia had PoderData 10–12 October with the candidates swapped.
3. **Tail-study verdict on the dashboard.** The label next to the published model in the comeback table said "within the defensible range" unconditionally. It is now computed (today: below the range, for the reason given in §C.4).
4. **Wikipedia scraper.** `fetch_polls_live.py` dropped every row with an end date after the first round as a "post-election poll", so no post-first-round runoff poll ever came from Wikipedia; all of them entered by hand. Fixed on 17 October: runoff rows count days to 25 October and are dropped only after it.

C.7 17 October

One new poll: Gerp, 14–16 October, n = 2,400, BR-05187/2026, Flávio 50, Lula 45, none 2, don't know 3 (two-party 52.63%). With 18 polls from 13 pollsters, $h = 8$: raw average 51.63%, corrected 52.94%, $\sigma_F = 0.096$, weights 0.92 / 0.08 / 0.01, $z = +0.18$; $\hat{s} = 52.9\%$ (48.2–57.5%), $\hat{\sigma} = 0.095$, Pr(Flávio) = 0.888 (t_4 tails: 0.920). Variants: latest round per pollster 0.864; fieldwork ending on or after 12 October 0.860; without AtlasIntel 0.878; without Gerp and Vox 0.880; Palver at effective n 0.894; undecided split evenly 0.887; drift-rule range for Pr(Lula) 0.103–0.112. Polls ending 5–10 October average 52.18% raw, those ending 11–16 October 51.28%.

C.6 Government approval

Approval is not in v7 (§5). For the record, the v6 input (mean net approval, good/excellent minus bad/terrible, of three pollsters) went from −8.5 to −7.5 (Datafolha 6–7 Oct, −5), then to −9.0 (AtlasIntel 3–8 Oct, −15.0), and stayed at −9.0 with Datafolha 13–14 Oct (−6) and Quaest 10–13 Oct (−6). v6, which also uses the polls, gives Flávio 0.493 on 16 October (0.480 on 6 October).

Working paper. Comments welcome.